



ASH DYKE MANAGEMENT Department
SIMHADRI

Note No :1



Ref:SMTTP/Ash Dyke Mgm/2024-25/NTPCSIMH/1093251

Date:

29/03/2025(11:22:25)

Subject:GUIDELINES FOR ASH DYKE MANAGEMENT LMI APPROVAL
REG.

As per the OGN ref. COS-IS0-00-OGN/OPS/SYST/022, LMI ref. LMI/OGN/OPS/SYST/022 was prepared.
The said LMI has been revised as per the revised OGN (Ver 4) released in March 2025.

Put up for approval please.

Submitted by:

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Note No:2

"GUIDELINES FOR ASH DYKE MANAGEMENT LMI "has been revised as per the revised OGN (Ver 4) released in March 2025.This has been checked and found OK.

Submitted for approval from Competent Authority

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Please review as discussed.

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Note No:4

Document updated based on discussion.

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Note No:6

Paragraph numbers may please be checked and revised from Page-54 to Page-61.

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Note No:7

Paragraph numbering has been revised from Pg. 54 to 61. Additionally, layout in Annexure-D has also been updated corresponding to Simhadri ash dyke.

Submitted for approval please.

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LMI is in line with the OGN. May please be approved.

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Note No:10

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NTPC SIMHADRI, GUIDELINES FOR ASH DYKE MANAGEMENT
LMI/OGN/OPS/SYST/022 REV. 5, March 2025

GUIDELINES FOR ASH DYKE MANAGEMENT

SIMHADRI SUPER THERMAL POWER STATION

**NTPC SIMHADRI, GUIDELINES FOR ASH DYKE
MANAGEMENT LMI/OGN/OPS/SYST/022 REV. 5,
March 2025**

LMI APPROVAL DOCUMENT

1	LMI NAME	GUIDELINES FOR ASH DYKE MANAGEMENT
2	LMI No.	LMI/OGN/OPS/SYST/022 REV. 5, March 2025
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4	PREPARED BY CHECKED BY	P C Gupta, Sr. Manager (ADM) Shri. Amrendra Kumar Raman, DGM (O&M-CIVIL)
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6	APPROVED BY	Shri. Sameer Sharma, ED (SIMHADRI)
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8	INDEX LIST	ENCLOSED
9	REVISION INDEX	ENCLOSED

**NTPC SIMHADRI, GUIDELINES FOR ASH DYKE
MANAGEMENT LMI/OGN/OPS/SYST/022 REV. 5,
March 2025**

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MANAGEMENT LMI/OGN/OPS/SYST/022 REV. 5,
March 2025**

REVISION INDEX
(For changes/revision)

Sl.No	Revision	Date	Details
1	3	28.05.2020	Instructions under different parts have been elaborated thoroughly
			Additional annexure's "Annexure-A Ash Dyke Capacity Calculations", "Annexure-B MOEF stipulations and Major Ash Utilisation areas" and "Annexure-E Schematic arrangement for sprinkler System" are included
2	4	08.06.2022	Revised in line with revised OGN "COS-ISO-00-OGN/OPS/SYST/022, Rev-3" released in May 2022.
3	5	27.03.2025	Revised in line with revised OGN "COS-ISO-00-OGN/OPS/SYST/022, Rev-4" released in March 2025.

Part-I General

1.0 Summary

1.1 Superseded Document: LMI/OGN/OPS/SYST/022 REV.4, June 2022

1.2 Scope : Guidelines for Ash Dyke Management in NTPC Stations

1.3 Review : Review Responsibility: Ash dyke management (Civil)

Review Frequency : 2 yearly basis, or as and when required,
whichever is less

Revision Approval : HoP Simhadri will be approving authority for
any revision in this document.

2.0 Objective

The objective of the guidelines is to ensure safety and stability of the dyke during planning, design, construction, operation and maintenance. These guidelines also provide for emergency preparedness. These guidelines are expected to make the people more safety conscious of the Ash Disposal Management. These guidelines recognizes that dyke safety is achieved through a continuous process, practices and procedures that shall be examined periodically and expedited with technological advancement for minimizing the risk of failure.

3.0 Record of Major Revisions:

- 1.0 Part-I, Cl. 4.0: Para added - The concerned Civil group has been made the nodal group.
- 2.0 Part-II: Deleted.
- 3.0 Part-III, Cl. 2.0: Sub Point g) and h) added.
- 4.0 Part-III, Cl. 2.1: Sub Point added.
- 5.0 Part-III, Cl. 4.1-i): Deleted.
- 6.0 Part-III, Cl. 4.1- iii) 3rd Para: Part addition at end as per OGN.
- 7.0 Part-III, Cl. 4.1-v): Deleted.
- 8.0 Part-III, Cl. 4.1-ix to xxi: Added as per OGN.
- 9.0 Part-III, Cl. 4.2-v & vi): Modified as per OGN.
- 10.0 Part-III, Cl. 4.2-vii): Deleted.

11.0 Part-III, Cl. 4.6 to 4.10): Revised as per OGN.

12.0 Annexure- H, J, K & L: Updated/added as per OGN.

4.0 Responsibility

The safety of the dyke has been of prime concern of the management that are evolved with various aspects of planning, design, construction, operation and ultimate disposal of ash. In the past, several guidelines were issued with the objective of ensuring that the most effective systems are established to ensure dyke safety. The safety of the ash disposal systems shall continue to be accorded highest consideration. Overlapping of ash filling and dyke raising activities in the same lagoon is to be avoided since this is a safety hazard to divide bund/ construction machinery/personnel working inside the lagoon and to main ash dyke itself. There is a need to identify responsibility Centre to coordinate with the concerned including planning/placing the award/completion of the construction/ raisings to ensure the adequacy of dyke safety program.

Each station shall prepare local management instructions (LMI) based upon this document fixing responsibility centres for the above-mentioned coordination, ash dyke monitoring and compliance of reporting procedures mentioned in this document.

An exclusive Ash Dyke Management (ADM) Department, comprising of three groups - Civil O&M, Ash Handling maintenance and Ash Utilization, was formed headed by senior executive at GM level in some of the stations. In remaining stations, however, O&M-Civil Department is continuing to look after the Civil-works in ash dyke area. The construction works involved in height raising, buttressing & strengthening of existing dykes are also executed by ADM / O&M-Civil departments at the respective stations.

The concerned Civil group, either in ADM department or in O&M-Civil department at respective stations shall act as the nodal group, do all planning, take the lead, coordinate with all concerned groups like, Ash Utilization, Ash Handling, EMG, Operation, Safety etc. to finalise actions with mutual consensus, take all preventive & corrective actions proactively wherever & whenever required and ensure safe & sustainable Operation & Maintenance of the ash dyke, including safe & timely execution of the aforesaid construction activities. The Ash Utilization & Ash Handling groups working in dyke area will align their activities accordingly, provide all support & assistance to the nodal group and act to achieve the common goal of ensuring safety & sustainability of ash dyke.

HOD (O&M) shall be overall responsible for Operation, Maintenance and Safety of the ash dyke.

5.0 Planning

Construction of dyke involves time to make ready for charging. In view of this,

the construction sequence shall be planned in such a way that there is no delay in availability / readiness of envisaged space for disposal of ash. Local management instruction (LMI) shall ensure to make the envisaged pond ready in all respect for discharge before the slurry level encroaches the free board in the pond(s) in service. The time process involved from FSC to award shall be reduced in order to reduce load on the dyke. Initiation of proposal, preparation of BOQ & specifications, issue of NIT for construction shall accordingly be scheduled to match with the ash generation, monsoon season, availability of sand etc. Whenever required, stations may club raising of 02 or more dykes in a single package to avoid multiple tendering process. With this, stations may be able to get better parties in view of larger jobs. This would also ensure that subsequent raising would be a continuous activity.

Rolling plan of ash dyke shall be prepared as stipulated under Part-III, Cl. 4.1-xvi.

As per latest notification dated 31.12.2021 “Any new as well as operational thermal power plant may be permitted an emergency or temporary ash pond with an area of 0.1 hectare per Mega Watt (MW)”. Even though ash is a good constructional material, however the avenues for use are seasonal and there may be some restriction by the local authorities on transport through public roads. In view of this intermittent usage of ash, adequate storage arrangement shall be planned.

6.0 Design

Deleted.

7.0 Construction

Construction is a critical phase in achieving a safe dyke. Design and construction, both shall be mutually reciprocated. Design shall envisage all the conditions of construction. The construction shall satisfy the assumed conditions in the design. Quality of construction is also critical to dyke safety. Defectiveness in quality of materials or construction practices can occur during any stage of the construction and efforts shall be made to prevent them.

Surveillance check for compaction shall be carried out by CC-OS-AMG in association with site representative.

Responsibility: HOD Ash dyke management (Civil)

8.0 Monitoring

Monitoring existing dykes and reacting quickly to inadequate performance or to danger signals is a continuing critical aspect for dyke safety. Careful monitoring and quick response can prevent failure largely. Instrumentation data will fore tell of any abnormal behaviour and quick response can prevent failures in many cases. These guidelines are intended to minimize failure in constructing new dykes and need to ensure the safety of existing dykes by inspection and maintenance and by adopting preventive measures and responding quickly in attending to any abnormal behaviour.

Responsibility: HOD Ash dyke management (Civil)

9.0 Inspection

The purpose of periodic inspection program is to verify throughout operating life of the project with structural integrity of the dyke and associated structures, assure protection to human life and property around the area. Periodic inspection will also disclose conditions, which may disrupt operation or threaten safety. Inspection shall be carried out as stipulated under Part-III, Cl. 4.2.

Responsibility: Cross functional team from Ash dyke management (Civil), EMG / Ash Utilization, Ash handling, Operation and MTP (if required)

10.0 Training

To supplement the personnel connected with operation, inspection, maintenance of the dyke, management shall provide training on regular basis. They shall be given broad exposure to variety of field conditions. Rotation of training programs at different sites shall be organized by PMI.

Responsibility: HOD (O&M)

11.0 Emergency Action Plan and Response Preparedness

In order to mitigate any probable failure, preventive measures shall be accorded top priority and each local management shall initiate measures to minimize the effects of failures. A suggestive emergency plan and response preparedness is included in these guidelines (Annexure-C), which shall be reworked as per local management instructions. Mock drills for ash dykes may be organized & conducted by station Safety Department in consultation with site HR as there is an involvement of local district administration. Report will also be prepared by

them and circulate to all concerned departments at station within a fortnight. Upon receipt of the report, concerned ADM/ O&M- Civil will share a copy to CC-OS-Ash Management Group. Any distress in the ash dyke is to be immediately reported to ED- Engg., CC-OS-AMG and RHQ. Critical materials such as sand/ash filled bags, boulders, gravels, filter media etc. may be made available near ash dyke to mitigate any emergency.

Responsibility: HOD (O&M)

12.0 Salient points

- i) Written operating instruction shall be prepared in the form of local management instructions (LMI).
- ii) The Cross Functional Teams formed for periodic inspection (refer Cl. Part-III, Cl.4.2 v) shall decide the ash slurry discharge points to ensure uniform disposal in the pond.
- iii) For natural filter materials, only grain size distribution shall be determined. But for artificial filters (geo-fabrics), elaborate testing shall be required.
- iv) A permanent approach to the water escape structure shall be provided at the design stage itself.
- v) Minimum free board as per design shall be maintained to avoid overtopping.
- vi) Periodic inspection shall be done by cross function groups of the project. Frequency of periodic inspection shall be 15 days in normal weather and 7 days or less during monsoon weather depending on dyke condition.
- vii) Sand chimney may be constructed up to 1.0m below the top of dyke.
- viii) Slope protection
 - a) On u/s slope if bricks are used, the same shall be dry flat brick lining inside panel walls.
 - b) Laboratory tests may be conducted on ash cement mortar before deciding the mix to be used.
- ix) The earth cover on the slopes shall necessarily be ensured. To prevent erosion, turfing shall be maintained. Geo textiles may also be considered for slope protection.
- x) Display of sign boards to prevent trespassing into Ash-pond area shall be installed at prominent places.
- xi) Steps to be provided on d/s slope at an interval of about 500m along the periphery

of the dyke.

xii) Sprinkling shall be done on continual basis.

xiii) Provision of separate collection and storage of bottom ash & fly ash shall be made, if not already available.

xiv) Inspection reports along with action points for major deviations of ash dykes as per Part – IV shall be sent every month to CC-OS-AMG, Project/ Station Engg. and Regional OS.

xv) Exception report shall be published in ORT each month.

xvi) CC-OS-AMG will organize audit of ash dykes once in a year that shall be carried out by a committee. However, depending on criticality of ash dyke, audit may be carried out on half yearly basis also. Audit shall cover the dyke safety also.

xvii) To minimize the risk of failures the preventive measures shall be accorded top priority. However, emergency action plan & response preparedness (EPRP) shall be kept in place. Mock drills shall be conducted at regular intervals as per Part-I / Cl. 11. Safety awareness shall be increased at each station.

xviii) To avoid erosion on u/s slope due to discharge points, the discharge points be extended inside the lagoon up to 5H from heel of dyke ('H' is the average height of dyke).

xix) From Safety point, provide suitable railing of about 1.5m height around WES wells above envisaged top level of wells. Well tops shall be covered suitably and a working platform all-round outer circumference shall be provided, along with a chain-pulley arrangement at slits for safe handling of precast slabs. Entry point at well approach shall be provided with gate arrangement for restricted entry. CC-Engg. shall provide drawings to station incorporating all such safety provisions.

Responsibility: HOD (O&M)

13.0 Recommendations:

i) Preventive measures shall be given top priority.

ii) Regular inspection shall be carried out by each station by cross functional groups. Exception reports shall be sent to CC -Ash Management group & Regional OS.

- iii) It is preferable to have a total plan for the design life of 25 years (with margin for extension by another 10 to 15 years) based on which short terms plans may be elaborated as per the requirement of the plant from time to time. Each station shall develop "dyke raising rolling plan for next 8-10 years and shall initiate action accordingly. All efforts shall be made to reduce the time in planning/placing the award for raisings. Any delay in planning/placing of award or completion can lead to overloading of dyke, which threaten the safety, and stability of dyke.
- iv) Monitoring of various Instruments shall be done regularly.
- v) There is a need to identify the responsibility center for planning, construction, operation, maintenance of the safety of the dyke during its operational life period.
- vi) A training in the Dyke safety and Management to be organized by PMI.
- vii) To have infrastructural support and manpower. An organizational chart shall be prepared for the Ash Disposal Management.
- viii) For testing of filter material, identification of reputed laboratories shall be done by regional headquarters. For natural filter materials, only grain size distribution shall be determined. But for artificial filters (geo-fabrics), elaborate testing shall be required.
- ix) Increased awareness of dyke safety and consequences of negative impact on environment is advisable.
- x) Annual Dyke safety Mock drill is to be conducted by each project as per Part-I / Cl. 11.
- xi) All Projects including JVs and subsidiaries are instructed to carry out Ash Dyke Stability Study compulsorily once for all the lagoons through external institutes of repute. Thereafter the study shall be carried out in the event of raising / strengthening or distress situation or abandonment, if any, for the lagoon concerned. The report should clearly indicate the remedial actions required to maintain the safety of the ash dykes.

Responsibility: HOD (O&M)

14.0 Main document

The document has been divided in 5 parts:

-Part I: General

-Part II: Deleted

-Part III: Guidelines for operation and maintenance of ash pond for storage of ash deposited by wet slurry method. They include Maintenance, frequency of inspection of dyke, Measures for avoiding overtopping, seepage points, rain cuts, discharging seepage water and measures to avoid water inundation. They also include Emergency action plan and response plan.

-Part IV: Guidelines for control of fugitive dust emission from ash ponds

-Part V: Check list of ash dyke inspection and maintenance.

-Annexures A to L.

The safety of the ash disposal systems shall continue to be accorded highest consideration. There is a need to identify responsibility centre to coordinate with the concerned including planning/placing the award/completion of the construction/raisings to ensure the adequacy of dyke safety program.

The guidelines are intended to outline the management procedures to use the best of Technology/practices that will continually improve methods in planning, design, construction, and operation to ensure safety of existing dykes and dykes to be constructed. These guidelines also provide for emergency preparedness.

15.0 REVIEW OF LMI

HOD Ash dyke management (Civil) shall review the LMI on two yearly basis.

Part-II Deleted

Part-III Guidelines for operation and maintenance of ash pond for storage of ash deposited by wet slurry method.

1.0 INTRODUCTION

This section of the guidelines outlines the activities for operation and maintenance, periodic inspection programs that which will ensure safe operation of the dyke. Preventive measures provide timely repair of dykes. An emergency action plan shall be kept in place so that the site management shall ensure to obtain safe operation of dyke. Overall responsibility for O&M and safety of ash dyke is fixed with Head of O&M. Written local management instruction (LMI) shall be prepared by each operating station. The instruction should cover the function of that dyke and describe procedures to follow to ensure safety of dyke.

2.0 METHOD OF SLURRY DISCHARGE:

- a) Initially the slurry shall be discharged near the water escape structure till deposition of ash near the WES. Once the ash deposition level around WES reaches about a meter below the spilling level, the discharge shall be shifted to other locations in following manner.
- b) Garlanding of header(s) shall be ensured for uniform ash deposition.
- c) Filling shall be done in such a manner that the ash deposition first takes place in the peripheral region just adjacent to the embankment all-round, followed by the central portion the lagoon. O&M team shall monitor the peripheral ash deposition level and after deposition in the coverage area of any particular discharge pipe, charging shall be switched to new location(s) through separate pipe(s).
- d) After peripheral filling, the discharge point(s) shall be extended towards central area of lagoon.
- e) Multi point discharge through multiple branches of suitable smaller diameter pipes from the headers shall be adopted to (i) achieve uniform ash filling within the lagoon, (ii) completely utilize the available storage capacity and (iii) maintenance of water cover throughout to avoid island formations of ash within the lagoon leading to fugitive dust problem.
- f) ADM team shall monitor & ensure that the level of ash deposition around WES shall not be abnormally low in comparison to that in rest of the lagoon, else, ash slurry will

rush towards the WES. This may be ensured by charging periodically around WES.

g) To avoid erosion on u/s slope due to discharge points, the discharge points to be extended inside the lagoon up to 5H from heel of dyke in Lean Slurry Disposal System ('H' is the avg. height of dyke). In other words, considering embankment formation height of 5m with 1V:3H side slope in any raising, the distance of discharge mouth shall be 43m away from the center line of the raising under charging [5x5m height + 3x5m (horizontal distance between heel & top upstream edge) + 3m (50% of top width of embankment)]. Measure for prevention of any such erosion are however already taken care in all new Starter dykes with provision of RCC chutes with energy dissipators in Engg. Drawings itself. No discharge shall be allowed on the slopes.

h) The Cross Functional Teams formed for periodic inspection (refer Cl. Part-III, Cl.4.2 v) shall decide the ash slurry discharge points to ensure uniform disposal in the pond.

2.1 Decanting System:

- TSS: Unless specified in consent letter from State Pollution Control Board otherwise, limits for TSS (Total Suspended Solids) in effluent shall be considered as 100 PPM to be measured weekly.
- For this purpose, samples shall be regularly collected and checked for TSS. Proper documentation shall be maintained with records of such measurements. In case the TSS is more than the permissible value sufficient, time for decantation shall be maintained. Responsibility: Chemistry
- For increasing the decantation time, increasing the height, of spillway may not be a practical solution always.
- Generally decanting wells are provided as Water Escape Structure for evacuation of clear water from main lagoons and spillways are provided additionally for emergency escape measure. As per present practice, WES wells are provided with flexible openings in the form of slits along with provision of precast slabs in Engg. drawings. Water level in lagoons may be controlled either by inserting or removing the precast slabs based on site requirement. No such provision is however kept in Spillways, where the spilling level is kept at predefined fixed RL.
- The practical way for efficient decantation is adequate lagoon size commensurate with the rate of inflow and efficient ash filling management (i.e. judicial shifting of

slurry discharge points)

- The flexible opening provided in the WES wells shall be used to a) prevent excess ponding and b) maintain adequate water cover over the deposited ash in the lagoon under charge to prevent any fugitive dust emission from storage lagoon.
- Every station shall comply to the actions suggested in the IOM, ref. COS/IOM/ED/22/064 date 16th Aug 2023, issued by ED (OS) (Annexure-K) to ensure healthiness of WES along with its flexible openings & to prevent ingress of ash / slurry into the wells, the decant pipes & the OFL.

Responsibility: Ash Dyke Inspection team

2.2 Maintenance of Ash Dyke

It is very important to constantly inspect the ash dyke and carryout necessary remedial measures wherever certain abnormal behaviour is observed/noticed. Following aspects must be considered during inspection of the dyke:

- Design slopes:** The slopes shall be maintained as per the drawings. In case of deficiency, make-up the slopes preferably by similar material.
- Top width:** The top width of the dyke shall be as per drawings. In case of any deficiency, the same shall be made up immediately.
- Top level of dyke:**
Top level of dyke shall be as mentioned in drawings. Any deficiency shall be made up immediately. Settlement/ sinking if any shall be immediately rectified by additional earth fill.
- Free board**
Free board as mentioned in the drawings shall be maintained. Measuring staff or Drone or any suitable device may be used.
- Earth cover and turfing**
Earth cover on the slopes mentioned in the drawings shall be maintained. Any erosion or deficiency in the slope, the same shall be covered with additional earth cover of 0.5 m compacted thickness immediately.
- u/s slope protection**
Slope protection shall be maintained as mentioned in drawings/specification.
- WBM road, Kerb /dowel, slope drains:**
These shall be as mentioned in the drawings. Defects if any shall be attended.
- Rock toe, toe drain, berm, rock pitching etc:**
Rock toe, toe drain, berm, rock pitching etc. shall be maintained as per the drawings. Dislodgements/defects if any shall be corrected.
- Instrumentation**
Piezometers surface settlement marker shall be maintained in working condition and protect from cattle crazing or theft. Defective instruments shall be replaced. Other details like spillway, WES etc design bed gradient of drains not specifically

mentioned but assist in maintaining the safety of dyke shall be attended as required.

- x) Deleted
- xi) From Safety point, provide suitable railing of about 1.5m height around WES wells above envisaged top level of wells. Well tops shall be covered suitably and a working platform all-round outer circumference shall be provided, along with a chain-pulley arrangement at slits for safe handling of precast slabs. Entry point at well approach shall be provided with gate arrangement for restricted entry. All such provisions shall be incorporated in drawings under guidance of Engg., if required.

Responsibility: Ash Dyke Management (Civil), AHM

2.2.1 Wet patches/softening on downstream slope

If the wet patches found to appear on the slopes, the area adjacent to downstream of the rock toe shall also be inspected. Corrective measures shall be taken immediately.

- a) Place reverse filter layer on wet patches extended by 5 metres all sides.
- b) Check ash deposition status inside lagoon on adjoining upstream of affected area and in case of ponding condition, develop ash deposition by careful charging under close observation.
- c) Providing relief wells, if required, near the affected area for safe exit of the seepage water.

Observe the condition and monitor the outflow from the relief wells. A register shall be maintained recording rate of flow from each relief well. Such measurements shall be taken at a frequency of 15 days. If any of the discharge pipes from the relief well is found to have been blocked, the same shall be cleared for effective relief of the seepage water.

Responsibility: Ash Dyke Management (Civil)

2.2.2 Gulley formation

Gulley formation on the downstream face due to surface water flow during rain can be prevented by maintaining grass turfing and by selecting non-erodible earth cover during the dyke construction. Further slope drains at intervals of 25-30 m will also help to avoid gulley formations. If any gully formation is noticed, the same shall be backfilled with cohesive soil and covered with grass turfing.

Responsibility: Ash Dyke Management (Civil)

2.2.3 Rat holes/animal burrows:

During inspection, if any rat holes or animal burrows are noticed, the same shall be plugged immediately using sand compacted by rod and then plug at the top with earth. The holes shall be filled and covered with grass turfing.

Responsibility: Ash Dyke Management (Civil)

2.2.4 Growth of plants:

Plants/trees shall not be allowed on the dyke top or slopes. Only shallow rooted grass shrubs can be allowed. If any plant growth is noticed, the area shall be cleared by removing all the roots, plug the area with selected soil and cover with grass turfing.

Responsibility: Ash Dyke Management (Civil)

2.2.5 Choking of surface drainage:

- i) Stagnation of seepage water in the drains is not desirable. This may be due to deposition of soil particles in the drain. All toe drains and surface drains shall be cleaned periodically by removing silt or vegetation for smooth flow. Design bed gradient shall be maintained.
- ii) Site/operation shall maintain record of total inflow into the ash dyke thorough various discharge pipes by suitable measuring system.
- iii) For routine maintenance works of ash dykes like repair of rain cuts/rat holes/gulley's/plugging of wet spots/cleaning of rock toe & peripheral drains, each station shall have Annual maintenance contract for ash dykes.

Responsibility: Ash Dyke Management (Civil)

2.2.6 Frequency of inspection

Refer Cl. 4.2 of Part-III - 'Inspection, Monitoring & Frequency'.

2.3 Preventive Measures:

2.3.1 Breach:

One of the main cause for breach of dyke/distress to the dyke is overtopping. To prevent overtopping, the design free board shall be ensured throughout the operating lagoons. Encroachment into free board must be avoided by advance planning of construction of the dyke raising in the other lagoons. All effort shall be made to,

- i) Ensure that future (next) lagoon shall be kept ready before the ash level reaches the maximum design fill level.
- ii) Free board of 1.5m shall be maintained.

2.3.2 Erosion and Slope Protection:

The erosion beneath / on the dyke slope is responsible for the subsidence and instability of the dyke. Some of the major causes are rain cuts, insufficient soil cover / turfing etc. Erosion on the u/s and d/s slope can be prevented by ensuring

- i) Compaction to the specified density both in the levelled portion of the dyke as well as on slopes.

- ii) Erosion of u/s slope of soil cover provided on ash embankment shall be prevented by pitching with flat brick / ash cement mortar lining preferably from the bed level or heel of the dyke up to the top level.
- iii) Erosion of downstream can be prevented by turfing and proper compaction of soil at sides besides providing slope drains to guide water from the dyke tops. Suitable geotextile may be used to avoid erosion of slopes.
- iv) By providing ripraps on the downstream wherever it is subjected to rainwater.
- v) All the rainwater from the dyke top shall be drained into the slope drains. These shall be inspected & maintained on regular basis.
- vi) Ensure proper gradient for the surface water to be drained through drains on the slopes.
- vii) Connection of dowel / kerb wall & drain is maintained for draining surface water from dyke top into the slope drains.
- viii) Trees shall not be allowed to grow on the slopes and on dyke tops.
- ix) It is necessary to fill all the gullies/cuts with earth and compact slopes and provide turfing much ahead of monsoon. Geotextiles may be provided for slope protection.

Responsibility: Ash Dyke Management (Civil)

2.3.3 Seepage and Piping

Piping or sand boiling is one of the main cause for excessive settlement or instability of the dyke. There are several reasons for this, however one of the main cause is improper drainage. This is primarily due to filter material not meeting the Filter criteria. Other causes may be due to use of oversized borrow material in the fill, insufficient cover or turfing, improper slope protection on the upstream slopes, encroachment into free board, rat holes etc. Improper connection between slope drains and toe drain, connection between the WES and Pipes or the construction joint in WES are also probable causes for seepage and piping. To prevent seepage and piping, the following shall be ensured:

- (a) The filter material shall satisfy the filter criteria.
- (b) Toe drain shall be clear of any blockages.
- (c) The invert of the toe drain should be below the blanket drain.
- (d) The discharge/ seepage water shall be monitored for its colours/ suspended particles.

- (e) The cross drainpipes from the toe drain shall be clear of all blockages.
- (f) The construction joints between old and new construction shall be properly keyed.
Responsibility: Ash Dyke Management (Civil)

2.3.4 Water Logging

In case of pervious foundation, the seepage from ponds can inundate the neighboring fields, causing damage to the fields. Water logging at the downstream shall be avoided to prevent subsidence/instability of the dyke. One of the main cause is improper connectivity from toe rain to natural drain.

To overcome this defect, a seepage barrier may be constructed around the dyke by means of a grout curtain or by any other suitable seepage barrier. Other suitable methods like artificial drainage through reverse sand wicks/relief wells, sand blanket surface loading etc., pumping arrangement may have to be provided. Intermittent sumps may also be considered in case of highly waterlogged areas. For details of relief wells refer IS 5050:1992.

Responsibility: Ash Dyke Management (Civil)

3.0 MONITORING THE DYKE:

Refer Cl. 4.2 of Part-III - 'Inspection, Monitoring & Frequency'.

4.0 OTHER GENERAL RECOMMENDATIONS

4.1 Effective Operation and Maintenance of Ash Dyke

Following are necessary for effective operation and maintenance of the ash dyke:

- i) Deleted.
- ii) Unauthorized entry into ash pond area shall be prohibited and display boards shall be installed at prominent locations. To avoid sabotage security guards for vigilance of the ash dyke area round the clock is preferable.
- iii) To facilitate inspection and maintenance the entire dyke perimeter shall have accessible roads with at least WBM topping. All around the dyke alignment there shall be a single lane inspection road at ground level, besides roads on dyke tops in all phases having provisions for overtaking zone of adequate width. At least one road (either at the ground level or at the starter dyke top) shall be bitumen covered and connected to a bitumen covered approach road in order to ensure an all- weather approach to ash dyke area. Ash dyke embankment should not be used for regular

plying of heavily loaded vehicles; however, exit/ entry of the vehicles into or from ash pond may done by Site by providing additional number of suitable approaches/ ramps on either side as per requirement and site condition. Wherever ramps are required as per site requirement to approach the dyke top, suitable drainage provision through number and size of pipes to be decided depending on discharge at toe drain shall be provided. At the junctions between approach road & dyke top, the kerb shall be replaced by suitable hump as per site conditions. Provision of guard wall/posts along edge on either side of ramp with fluorescent white & black colours stripe paints shall also be made. PCC saucer drain may be constructed on either side of ramp to avoid damage to ramp from rainwater.

- iv) Flood light masts and CC Cameras on the dyke area as required may be provided for area illumination and monitoring. These lights need to be turned off during daytime.
- v) A site office/ porta cabin may be considered at the ash pond area. The same may be provided with various facilities like, telephone, PC system with LAN connection, security etc.
- vi) One dedicated Vehicle should be allotted for ash dyke ADM/O&M(Civil) group. One additional vehicle may also be deployed during monsoon seasons depending on requirement.
- vii) Routine Maintenance Contract shall be in place on Annual / Biennial basis. Stage wise separate contracts may be preferred.
- viii) Separate Annual / Biennial contract shall be in place for perennial 'Patrolling, Inspection & Assistance' round the clock through deployment of adequate manpower equipped with mobile phone so that they can promptly report any untoward development to concerned ADM / O&M executives. The deployed manpower may also be utilized for attending immediate remedial measures in such events.
- ix) Details of topographical survey, earlier land use map, geotechnical investigation report, location of ground water area drainage study report shall be available with ADM/ O&M Civil, besides TS or Achieves.
- x) Coal boulders option may be considered in lieu of stone boulder pitching wherever possible provided it meets the requirements of material properties.

- xi) Bottom ash shall be used in place of sand, as filter material wherever it meets the filter criteria. For this, storage of Bottom ash in separate lagoon is required for using bottom ash as filter material in raising ash dyke. Provision of separate collection, transportation and storage of bottom ash & fly ash shall be explored, wherever not available. This will also enable making ash available as per users' requirement.
- xii) Provision of RCC perimeter drain all along the perimeter of dyke parallel to inspection road (WBM or Bituminous) shall be provided to collect the seepage water from toe drain to guide seepage water up to Seepage water pump house. Arrangements shall be made to connect toe drain to perimeter drain at lowest point so that all accumulated water may flow into perimeter drain.
- xiii) Safety: Only qualified and trained personnel should work at the storage area. Necessary safety protection device should be provided to the personnel. One person alone must never enter a slurry compartment and should always be accompanied by a second person and both persons should wear necessary safety gear.
- xiv) Rehabilitation of the storage area: After completion of the storage area operations, designated areas can be rehabilitated by covering these areas by topsoil and vegetation after checking stability and consistency. It is advised that a civil engineering survey is to be executed to verify the stability of the filled area.
- xv) Steps shall be provided at interval of 500 meters c/c along the dyke on D/s slope of the embankment for inspection purpose.
- xvi) Long Term Rolling Plan: Each station shall prepare a long term (for 8-10 years) rolling plan of ash dyke incorporating space available in present phase, ongoing process of new space creation, envisaged future space including new ash dykes (if any) in format already shared by CC-OS-AMG considering yearly ash generation and anticipated ash utilization to have enough capacity for ash disposal without encroaching free board. Necessary clearance / NOC, as applicable, for any raising / buttressing / capacity addition shall be obtained from SPCB / appropriate authority well in advance. The plan shall be reviewed/updated half yearly or earlier, if required. A copy of the plan shall be shared with CC-OS-AMG.

In-situ dry density of pond ash varies from 0.78 to 1.05 gm /cc, specific gravity varies from 1.58 to 2.21, and average density of 1.0 tons / m³ may be taken for storage volume calculations in wet disposal system.

xvii) To have infrastructural support and manpower. an organizational chart shall be prepared for the Ash Disposal Management.

xviii) Filter material, coarse aggregate, rock/stone:

The material for filter shall meet the filter criteria as mentioned in the relevant technical specification. Material for coarse aggregate and stone/rock for toe drain/rock toe/slope protection shall meet the requirements as mentioned in specification. Necessary tests as per FQP shall be carried out from reputed laboratory as approved by the owner.

For testing of filter material, identification of reputed laboratories shall be done by regional headquarters. For natural filter materials, only grain size distribution shall be determined. But for artificial filters (geo-fabrics), elaborate testing shall be required.

xix) Chainage pillars with marking shall be provided at every 100 meters along periphery of dyke in each phase for reference.

xx) Realistic assessment of periodic Residual Capacity: Field survey shall be carried out by ADM/O&M-C group before charging in any lagoon in each phase for assessment of actual space created. In case of Starter Dyke, the same shall be carried out jointly by CCD & O&M with proper documentation. Residual space at the end of each subsequent month shall be assessed based on the above figure, considering the quantity of ash charged and lagoon-wise void created, if any, through utilization of pond ash during the month. Operation & AHD (Dyke group) shall maintain record of lagoon-wise total inflow into the ash dyke thorough various discharge pipes by suitable measuring system and provide the data to ADM/O&M-C group. The volume of void created in each lagoon during the month through utilization of pond ash shall be provided by AUD. ADM/O&M-C group will prepare monthly status report showing the lagoon-wise residual space availability in format attached as Annexure-H & share a copy to COS-AMG by the first day of following month.

xxi) Water level in all the lagoons including OFL shall be monitored regularly at WES

wells & spillways. Graduated scales or any other suitable measuring arrangement shall be adopted for this purpose.

Responsibility: Ash Dyke Management (Civil), EMG (AU)

4.2 Inspection, Monitoring & Frequency

- i) The pond area which is under operation shall be inspected on regular basis by site ADM / O&M- Civil team. If any subsidence, sink holes or craters are noticed on the surface of pond, the downstream side at the same location shall be inspected for any other signs of weakness/abnormality in the nearby area. Immediate preventive measures shall be taken. A site report of such observations shall be recorded &, if necessary, be communicated to Station Engg. & CC-OS-AMG.
- ii) Use of new technology like Drone & AI for periodic inspection may be adopted.
- iii) All the operating and non-operating dykes shall be inspected practically after events like earthquake, cyclone, heavy rains, high flood in the river, etc. and a report prepared based on the observation. If any damage is noticed, the same shall be rectified as per the guidelines given in this note. If no suitable guidelines are found for the nature of the damage, Station Engg. & CC-OS-AMG shall be consulted immediately.
- iv) To confirm the performance of the dyke as per the design requirements, it is essential to monitor the performance of the dyke throughout its operation. Instruments commonly provided for such monitoring are listed below:
 - a) Settlement gauges along the top and slopes of dyke.
 - b) Piezometers, minimum 3 to 4 nos. at critical sections to check the phreatic line during various stages of operation to verify the efficiency of internal drains.
 - c) Suitable device shall be installed at convenient location for Monitoring of free board.

All the instruments for monitoring purpose mentioned above shall be protected against damage by the local people and by movement by vehicles and cattle. The measuring instruments shall be kept under safe custody and regularly cleaned to

prevent corrosion and malfunctioning. The batteries if any shall be regularly charged or replaced. Readings of piezometers and settlement shall be recorded on monthly basis in the format provided in Annexure-E appended at the end of this guideline. Frequency may be increased during monsoon period and in case of any distress and proper records to be maintained.

v) Every station shall form a Cross Functional Team consisting of site executives from ADM / Civil (O&M or Construction), EMG / Ash Management, Ash handling, Operation and MTP (if required) for carrying out Inspection once in 15 days during normal season and during monsoon once in 7 days or less than 7 days depending on dyke condition. Convener of the inspection team will be Site ADM / Civil (O&M or Construction) executive.

The periodic inspection shall be carried out exclusively using existing NTPC mobile app for ash dyke inspection, inbuilt with the checklists for inspection, as provided in Part-V of this guideline for inspection. The reports will get generated upon concluding the inspection and will be delivered through auto generated mails to concerned CFT members & HOP, with copies to Station Engg & COS-AMG. However, station will send the reports to Regional OS. The inspection report is to be annexed to ORT report every month. Action points with targets for major deviations captured during inspection shall be sent every month to CC-OS-AMG, Station Engg. and Regional OS. System will also send auto generated reminder mails to concerned HOP with copies to Station Engg. & COS- AMG in the event of any delay in maintaining frequency of inspection vis-a-vis the stipulated frequency. The mobile app will be updated periodically to address issues, if any.

All exceptions observed during inspection by site level committee on ash dyke will be included in the exception report prepared by ORT coordinator every month. ORT coordinator shall also record free board level of ash dykes in the ORT format (as per Annexure - B) pertaining to Ash dyke.

Regional OS shall prepare an exception report for the stations falling within the region with regard to ash dyke safety, environmental conditions and dyke raising

status. The same shall be put up to RED and a copy shall be sent to CC-OS-AMG and Station Engineering. CC-OS-AMG shall compile the exception reports received from regional OS and put up the exception reports of all the stations to Director (O), ED (OS) and ED (Engg).

vi) Generally, the frequency of Audit of ash dykes will be biennial in line with that of the Technical Audits. However, the frequency will be annual for station(s) having specific issue(s) / concern(s), if any, either prevailing or newly developed, related to healthiness of dyke or space crisis. CC-OS- AMG will constitute a three-member committee to conduct the audit, with participants from CC- OS-AMG or any associate of CC-OS-AMG, along with executives with Civil background from ADM/O&M/Erection from any station(s) in any other region(s) and/or any executive from concerned RHQ, and/ or any executive with Civil background from Station Engineering group. Committee report shall be put up by CC-OS-AMG for approval of CA and approved document will be shared with all concerned. ROS ORT coordinator shall monitor compliance status of audit action points and attach the findings with ORT report.

vii) Deleted.

viii) Routine monitoring of OFL water level helps in avoiding any surprise of abnormal surge in ponding level & its probable consequences. For this graduated scale shall be painted on side wall of emergency spillway. Likewise, installing graduated scale near inlet mouth of AWRS pipes will give prior indications about abnormal high or low ponding level for efficient operation of AWRS pumps accordingly. Frequency of monitoring may be once in a month or lesser depending on site conditions. Any abnormality shall be reported to head of O&M.

ix) Monthly status Report is to be generated with exceptions & shared with CC-OS-AMG as per suggested format in Annexure-H.

Responsibility: Ash Dyke Management (Civil), Ash Dyke Inspection team

4.3 Manpower Requirement:

Site management shall ensure adequate and complete staff to perform its functions

in operation, inspection, and maintenance of dyke safety. It is essential that support personnel and equipment/facilities are provided to accommodate the needed maintenance activities.

4.4 Emergency action Plan:

Despite the guidelines in design, construction, operation and maintenance and inspection, the possibility of dyke failure small or big may not be completely ruled out. Pre-planning is required to identify condition which could lead to failure. In order to prevent failures, measures shall be in place to minimize the effects of such failures.

To minimize the risk of failures the preventive measures shall be adopted on top priority. However, emergency action plan & response preparedness (EPRP) shall be kept in place. Mock drills shall be conducted once in a year as per Part-I / Cl. 11. Safety awareness shall be increased at each station. The Emergency Preparedness plan is given in Annexure C.

Each station to report major area of concern of ash dykes and its action plan to tackle the situation is to be given to CC-OS- Ash Management group, Project/station Engg. And RHQ.

4.5 Monsoon Preparedness:

- i) To handle emergency situations during monsoons provision for deployment of adequate equipment like excavator / JCB, tipper, utility vehicle, DG sets, dewatering pumps etc. along with unskilled manpower shall be kept in routine maintenance contract or through separate contract.
- ii) As part of emergency preparedness, sufficient nos. of 50 kg bags (empty cement bags) filled with ash/ sand / earth, balla, stone aggregate, sand / Bottom Ash, stone boulder, earth, empty 50 kg bags, Balloon Lights etc. should be stacked at probable weak locations before onset of monsoon.
- iii) ADM / O&M-Civil group shall attend all checklist points for monsoon preparedness well in advance of onset of monsoon and furnish it to head of O&M, HOP, RHQ & CC-OS-AMG by the month of May. The suggested checklist has been appended in Annexure-F.
- iv) Concerned dyke O&M team may be equipped with torch, gumboot, umbrella / raincoats etc. to facilitate smooth inspections & maintenance during monsoons /

inclement weather conditions.

v) During Monsoon seasons, all critical ash dykes should be kept under close inspection & monitoring during night / odd hours. Team of senior level O&M executives shall be deputed on daily basis during monsoon season for such inspection through rota system. The team will submit their observation report to head of O&M latest by the following morning. Major exceptions, if any, shall be promptly brought in the knowledge of Station Engg. and CC-OS-AMG by station.

Responsibility: Ash Dyke Management (Civil), AHM

4.6 Method of Ash Evacuation for utilization:

To ensure dyke stability & safety, ash evacuation shall start from not less than 5H (five times the height of the embankment) measured from the upstream slope surface meeting the ash deposition level at freeboard level (Ash deposition inside lagoon is not permitted above freeboard level) with side slope of evacuation pit same as that of the embankment, i.e. 1V:3H. For more clarity & detail stipulations, Approved Engg. Guidelines for recovering land from existing Ash Ponds of NTPC Stations, ref. EOC/PE CIVIL/2019-20/ASH POND/856371 dtd. 04.02.2020 (Annexure-I) shall be followed. The evacuation start line shall be marked using flag posts beforehand. Movement of heavy earth moving vehicles on the top of the embankments shall be avoided as it may lead to safety hazards to the embankment, like, damage to the designed filter media, bulging out of slopes, longitudinal cracks etc., as the dyke sections are neither designed for such movements nor such design may be economical. Only crossing of dyke top road may be provided at suitable locations with required ramps, wherever peripheral road is available on downstream of starter dyke toe & CC-Engg. will provide such assistance whenever sought by station. In case of non-availability of such peripheral road, ash evacuation may start further away, over & above that stipulated in Annexure-I, to accommodate & create a peripheral haul road of adequate width over ash deposition inside lagoon.

Ash evacuation beyond the depth mentioned in the afore mentioned Engg. Guideline dated 04.02.2020, is not recommended. However, if required, same can be evaluated & advised suitably by Station Engg. On case-to-case basis on specific request of Station(s).

Every station shall comply to the actions suggested in the IOM, ref. COS/ED/AMG/2024/53 date 9th Dec 2024, issued by ED (OS) (Annexure-L) to ensure safety of men & equipment involved in heavy vehicle movement at Ash Dyke Area.

Responsibility: EMG (AU)

4.7 Stability Study (software-based analysis):

All Projects including JVs and subsidiaries are instructed to carry out software-based analysis for Stability Study of Ash Dyke compulsorily once for all the lagoons through external institutes of repute. Thereafter the study shall be carried out in the event of raising / strengthening or distress situation or abandonment, if any, for the lagoon concerned. The report should clearly indicate the remedial actions required to maintain the safety of the ash dykes.

Responsibility: Ash Dyke Management (Civil)

4.8 ANNUAL CERTIFICATION OF ASH PONDS AND DYKES:

A. Background:

In line with MoEF&CC Ash Notification dated 31.12.2021 (as amended by amendment notification 30.12.2022), CPCB & CEA have issued 'Guidelines on Design, Construction, O&M and Annual Certification of Coal Ash Ponds' in June 2023. As per the stipulation under Chapter 9 of the guideline, the coal or lignite based thermal power plants will have to ensure Annual Certification of the operational as well as stabilized and reclaimed ash ponds and dykes by a qualified professional geotechnical engineer and shall submit annual implementation report about the compliance of provisions in the notification by the 30th day of April, every year to CPCB and concerned SPCB or Pollution Control Committee (PCC), CEA, and concerned Integrated Regional Office of MoEF&CC.

Accordingly, all NTPC / JV stations having Ash ponds / Dykes will have to ensure to get the annual certification done and to submit the report to the authorities in line with the said guidelines every year starting from current financial year. Intimation about compliance to it will be given by the stations to COS- AMG, on or before 30th day of April, every year, along with the copy of the report.

The above mentioned CPCB & CEA guideline can be accessed using link:
https://cea.nic.in/wp-content/uploads/tcd/2023/06/GUIDELINES_ON_DESIGN_CONSTRUCTION_OM.pdf

B. Roles defined in the guideline of the certifying expert:

Under Cl. 9.1 (5) & (6) of the CPCB & CEA guideline, the details have been provided on which the certifying expert will have to submit his report. A checklist for the annual certification has been provided under Cl. 9.3.

The certifying expert will ask for various documentations, like, construction drawings, quality control documents, periodic inspection reports, audit reports, instrument records, TSS of decant Water, Annual Audit report of Ash Generation & Utilization, any other relevant reports etc.

The expert will also visit site to carry out physical inspections related to safety of the ash dyke, including taking certain site measurements / readings, along with other aspects like, Mode of ash disposal, water consumption or conservation in disposal, ash water recycling, Environmental pollution, green belt etc.

Based on the inputs gathered, as above, the certifying expert is likely to correlate & draw interpretations while certifying the ash pond.

C. Actions required to be taken by all stations:

Various actions to be taken on time by all stations include the following:

- i) Ash Dyke Management / Plant Civil O&M (Ash Dyke Group) of each station shall be the nodal department for the certification process. They will undertake & execute the activities & formalities involved and get the certification done including submission of the same to the authorities.
- ii) Station EMG will provide all relevant information & records and get associated with the nodal department during certification process in aspects pertaining to the environment. Station Chemistry Department will provide records of TSS of decant water, the Ash Utilization department (wherever not included in ADM) will provide information / relevant documents in support of 'Check list for Fly Ash generation and utilization' (Cl. 9.1-5-j of CPCB & CEA guideline). Station EEMG will provide water consumption (checklist point no. 10 under Cl. 9.3) in wet disposal to ash dyke.
- iii) Engagement of a qualified professional geotechnical engineer, from any reputed Govt. institute, with scope of work defined clearly in the contract along with definite timeline for report submission in compliance to the guideline.
- iv) The nodal department at each station shall perennially maintain all relevant reports & documentations in chronological order and shall make them available promptly when asked for by the expert.
- v) As per Cl. 9.1-5(a), read in combination with checklist point no. 24 (1) under Cl. 9.3 of the CPCB & CEA guideline, the dyke structural stability is to be examined by the certifying expert based on the construction drawings, quality control documents, monitoring reports etc.
- vi) Environmental points for Annual Certification shall be established based on

following:

- a. 100% recirculation of ash pond overflow and underflow: Ash Water Recirculation system (AWRS) and Toe Drain Water Recirculation System (TDWRS) are in place and in operation.
- b. Fugitive Dust Control: Fugitive dust control measures like Water cover, water sprinkling network, vegetation cover, earth cover etc.
- c. During transportation of ash, the truck mounted water sprinklers are used in haul roads and Ash Vehicles are covered with Tarpaulin before leaving from the ash dyke to be ensured.
- d. Vehicle Tyre washing system may be provided at suitable locations.
- e. Plantation around the ash dyke for fugitive dust emission control.
- f. Manual Sampling of Ambient Air quality and water quality – Frequency: Monthly once.

4.9 Requirement of Toe drain/ Seepage water pump house:

NGT & SPCBs have issued directive for zero discharge from Plant and prohibiting any discharge of liquid effluent from ash dyke into the river or any surface water body. Accordingly, the toe drain water from the ash dykes is not to be discharged into the nearest river/ water body and needs to be recirculated. In recent projects, the above aspect is taken care during the engineering of the AHS/ Ash dyke package where in toe drain water is envisaged to be pumped back into ash dyke for recirculation. However, in older projects where this scheme is not provided in the original drawings, a “Guideline for finalizing & implementation of pumping arrangement for recirculation of toe drain water back to the ash dyke” dated 09.09.16 has been issued. Site to ensure that system of recirculation of the toe drain water is implemented. All efforts shall be made to avoid waterlogging in and around the dyke.

4.10 Schematic sketches have been annexed in Annexure-J, showing typical lagoon arrangement in Ash Dykes without HCSD and typical water escape structure.

Part-IV GUIDELINES FOR CONTROL OF FUGITIVE DUST FROM ASH PONDS

1.0 Operative Lagoons:

1.1 Ponding of Water and Maintenance of Freeboard:

In the operative lagoon, ponding of water is essential in the operating lagoon to ensure proper sedimentation of ash particles. The water cover will also help in spreading ash deposition to farther distances from inlet points towards water escape structure. In case of any non-uniform deposition of ash along the periphery of the dyke due to insufficient number of discharge points, flexible pipes of HDPE etc. may be utilised for uniform ash deposition, in the ash deficient pockets. The discharge points may be shifted before the ash emerges out of the design water surface. The lighter and flexible pipes may also be used for the farther extensions, deep into the lagoon. Maintenance of water cover will also eliminate fugitive dust emission from the ash lagoons. A free board of 1.5m from the top of the dyke is kept from various design considerations. As per MOEF guidelines, the operative lagoon is designed in such a way that decant water, before escaping, should cover entire ash surface in the lagoon, by ponding and by maintaining the design free board.

By maintaining the freeboard as mentioned above, the ash settles uniformly under water and till the time ash deposition level is below the design level, i.e., below the overflow level of the water escape structure, the dust emission from the operative lagoon will be under control and can be eliminated by ensuring water cover in the entire lagoon area.

Fugitive emission can also be controlled by regulating water level using precast slabs in flexible opening of WES.

2.0 NON-OPERATIVE LAGOON:

In the non-operative lagoon, for the construction of dyke and borrowing ash, the top layer of the ash bed for a depth of about 1.0m to 2.0m, has to be dry so that equipment/machinery can move over the ash bed for the raising activities, viz, ash excavation in the borrow area, transportation and dyke construction. The dry ash, from the surface of the non-operative lagoon, normally flies during strong winds in summer season. The ash flying is more acute locally in the borrow areas due to these raising activities.

2.1 Rolling and Compaction:

The construction activities for raising of dykes and borrowing of ash will usually be going on near the periphery. Hence in the central region of the lagoon and some of the peripheral areas with no construction activities, the ash surface may be rolled and compacted which will reasonably stop the ash particles being lifted away by the mild

wind flow over the surface. Rolling and compaction of ash surface with occasional wetting as per requirement, is techno-economically a better option for such undisturbed areas. However, during strong summer winds, the dust emission can be controlled, in such areas, through occasional and controlled slurry/water discharge, so that the ash surface is kept just wet. In such areas, covering the ash surface with thin layer of earth is also an alternative solution.

2.2 Slow/Occasional Discharge of Water/slurry to Wet Ash Surface:

Even in the non-operative lagoon, had the ash filling been stopped before the freeboard of 1.5m is encroached, the dust emission from the ash dyke could be minimized by occasional slurry/water discharge, at a low discharge rate (i.e., by slow filling} into the non-operative lagoon to just wet the ash surface. The non-operative lagoon shall not be kept dry for long time and raising to be started at the earliest. Till such time, slow filling to be resorted to. And since the ash surface will be below the designed level, major area of the ash surface can easily be made wet. In case, local ponding is required in some areas, small ponds may be created using small height ash bunds confined with empty cement bags filled with ash.

For the borrow area, used for borrowing ash for dyke construction during ash dyke raising, the dust emission, may be controlled locally by sprinklers, etc and these measures in the borrow areas may also be suitably included in the scope of the contractor, for dyke raising.

2.3 Sprinkling System:

Each station shall prepare scheme for controlling fugitive dust emission. If spray is to be done, technical specifications, pipeline layout shall be prepared by station FES based upon typical schematic of sprinkler system as given in this document.

The lagoons especially on the perimeter, where the fugitive dust is severe, swivel type sprinklers are to be provided at every 50 metre. The sprinkler size shall be minimum 65mm. For providing water to the sprinklers, following arrangement is required:

A permanent pump house is to be provided near overflow lagoon having adequate pumping capacity and head to meet requirement. (Site may contact station Engg. To get detailed scheme if required). The water from overflow lagoon shall be used for sprinkling purpose.

Temporary sumps on the ash pond, where the slurry is not being discharged, shall be constructed out of brick masonry of 10 m dia. and 1.5 m depth, in which the water shall be pumped. The size of the sump may vary depending on the site requirement.

Near these sumps, pumps of adequate pumping capacity and head may be installed to meet sprinkling requirement.

The pipe network shall be made using used ash disposal pipes.

Power supply for the pumps, may be taken from the nearby source (viz, from NTPC or from the State Government}, whichever is techno economical.

A typical sketch showing the schematic arrangement for the sprinkling system has been placed at Annexure-D.

Flexible portable Sprinkling system with suitable pumps may also be adopted as a convenient dust suppression technique from embankments under raising, inoperative lagoons & also on localized high deposition of ash in operative lagoons.

2.4 Use of lean cement and ash slurry sprinkling:

By laying a very thin layer of lean cement ash slurry (4% -6% cement and 94 -96% ash) by sprinkling system can also be used for control of fugitive dust in non-operative ash dykes.

2.5 Use of HDPE sheets:

Covering the ash surface with thick HDPE sheets and weighted with stone boulders at intervals.

Responsibility: Ash Dyke Management (Civil), AHM

3.0 PRECAUTIONS FOR OVERFILLING TENDENCY:

It has been observed that to utilise the lagoon storage capacity to the maximum possible extent, some of the project sites tend to overfill the ash ponds up to levels more than the design fill levels. In this regard, it may kindly be noted that during intermediate stages for raisings, overfilling does not enhance the overall storage capacity of the ponds, except that it creates problems of ash flying, as the ash emerges out in the free board area above the design water level. These higher spots, where the ash has emerged above the water surface, contribute to the dust emission, even in the operative lagoon and it will be difficult to wet these higher spots subsequently. Further, in non-operative overfilled pond, there is always a danger of rainwater flowing from the higher ash filled area towards the dyke bund during heavy downpours and in the process, may result in the failure of dyke due to overtopping. A free board of 1.5m is normally kept to avoid all such problems and accordingly the invert level of water escape structure is also kept at 1.5m below the top of the dyke, called the designed ash fill level. It is, therefore, advisable not to overfill the dykes and to take immediate measures for the areas already overfilled (if any), by suitably guiding the water towards the water escape structure (i.e., away from the peripheral dyke) to stop the rain water from over-topping the dyke.

Responsibility: Ash Dyke Inspection team

4.0 Other measures:

Various other measures like, creating green cover by plantation of shrubs like beshram, all-round green curtain, especially useful to prevent emission during ash evacuation, water-tankers fitted with pumps & long flexible delivery hose, rain guns etc. are also effective for controlling fugitive emission

5.0 Sample Format for measures to control Fugitive emission:

Dust emission aggravates with wind blows during Summer and in dry winter seasons. A checklist of action taken for implementation of dust suppression measures shall be prepared by all stations well in advance and furnish it to head of O&M, HOP, RHQ & CC-OS-AMG. The sample format has been appended in Annexure-G.

Part-V Checklist for Ash Dyke Inspection

The checklist points have already been inbuilt in NTPC mobile app for ash dyke inspection (Refer Clause No. 4.2-v of Part-III)

Name of the Station	
Date of inspection	
Name of the inspecting officers	
Designation	

	ASH DYKE No./NAME			
INSPECTION DETAILS	1	2	3	REMARK
A. ASH LAGOON DETAILS				
1. Water level in the operating lagoon (RL)				
2. Available Free board (meters)				
3. Whether any ash surface is exposed abovewater.				
4. If ash surface is exposed above water level whether ash is flying anywhere.				
5. Whether water is flowing through all the openings of the water escape structure.				
(a) Approximate head of water flow over the well slabs. (b) Approximate depth of water over ash surface around/ near operating water escape structure.				
6. Whether any inter slab leakages of ash are present in the water escape wells.				

7. Whether water flow is obstructed by floating plants or any other floating bodies near the vents in the water escape structure.				
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INSPECTION DETAILS	ASH DYKE No./NAME			REMARK
	1	2	3	
B DYKE DETAILS				
1. Top level of dyke (RL)				
2. Nos. of raisings done.				
3. Whether there is any sign of settlement on the top of dyke.				
4. Whether there is any signs of sinking/ caving-in/boiling on the (a) upstream (b) downstream (c) Foundation very near to the downstream toe.				
5. Whether any wet spots/ areas are observed on (a) Downstream slope & (b) Foundation very near to the downstream toe.				
6. Whether any seepage is observed on (a) Upstream slopes (b) Downstream slope (c) On the foundation very near to the downstream toe.				
7. Whether any longitudinal cracks are observed (a) On the top of dyke (b) Upstream slope (c) Downstream toe.				
8. Whether any transverse cracks are observed (a) On the top of dyke (b) Upstream slope (c) Downstream toe.				

9. If transverse cracks are observed on the top and the slope				
- Whether the cracks are continuous.				
- Whether the length of cracks is increasing with time.				
- Whether the cracks are widening with time.				

	ASH DYKE No./NAME			
INSPECTION DETAILS	1	2	3	REMARK
10. If seepage is observed on the slope or near the downstream toe whether the seepage water is clear/ muddy indicating material carry-over.				
11. If the seepage water is muddy, TSS level at that point.				
12. If filters have been placed over the seepage area, whether the water have become clear indicating reduction in material carry over.				
13. Whether the seepage rate is changing with time-increasing/ decreasing/ constant.				
14. Whether the filter materials as mentioned above are getting displaced due to seepage water flow.				
15. Whether there is any gap/settlement between the effluent draw-off pipe and the surrounding dyke material visible from the downstream side of the dyke.				
16. Whether there is any erosion on the upstream slope due to wave action or draw down in the lagoon.				
17. Whether there is any damage to the turfing protection on downstream slope.				
18. Whether any stone pitching/ brick lining is dislodged or caved in on				
(a) Upstream slope				
(b) Downstream toe.				

19. Whether there is any growth of vegetation/ bushes (a) On the top of dyke (b) Upstream slope (c) Downstream toe.				
20. Whether any rat holes are present on the top of dyke, upstream & downstream slope.				
	ASH DYKE No./NAME			
INSPECTION DETAILS	1	2	3	REMARK
21. If rat holes are present, whether they are being plugged.				
22. If rat holes are present, whether there are also signs of cracking, sinking or settlement on (a) The tope of dyke (b) Upstream slope (c) Downstream toe (d) Near the region where rat holes are found.				
23. Whether there are any rain cuts/gullies on (a) top of dyke (b) upstream slope (c) downstream slope.				
24. Whether the rock toe is maintaining its design shape.				
25. Whether the lining in the toe is in good condition				
26. Whether there is any flow in the toe drain.				
27. Whether the toe drain is clean with no obstruction for flow of water.				

28. If no, (a) whether the toe drain is filled with dykematerial (b) whether the design downstream slopeof the dyke is maintained (slope to be always maintained)				
29. Whether the toe drain is shifted/ slided from itsoriginal position.				
30. Whether there is any blockage in the cross pipes provided in the drain.				

Annexure-A

Salient MOEF Stipulations and Major Ash Utilization Areas:

1. As per latest MoEF latest Gazette notification dated 31.12.2021, every coal or lignite based thermal power plant shall be responsible to utilise 100 per cent ash (fly ash and bottom ash) generated during that year, however, in no case shall utilisation fall below 80 per cent in any year, and the thermal power plant shall achieve average ash utilisation of 100 per cent in a three years cycle: Provided that the three years cycle applicable for the first time is extendable by one year for the thermal power plants where ash utilisation is in the range of 60-80 per cent, and two years where ash utilisation is below 60 per cent and for the purpose of calculation of percentage of ash utilisation.

Above notification further states, "Any new as well as operational thermal power plant may be permitted an emergency or temporary ash pond with an area of 0.1 hectare per Mega Watt (MW)".

2. Utilization avenues of Ash based on Pozzolanic Properties:
 - a. Manufacture of Portland Pozzolana Cement
 - b. As part replacement of cement in cement concrete and mortar.
 - c. Manufacture of Ash based building products like Ash Bricks /Blocks/Tiles, Aerated
 - d. concrete products, Lightweight Aggregates etc.
 - e. Controlled Low Strength Material (CLSM) : Used as fill material.
3. Utilization avenues of Ash as Inert Material:
 - a. Construction of road/ rail embankments, dykes, landfills, road sub-base and base course
 - b. Stowing in underground mines
 - c. Reclamation of decoaled areas of open cast mines.
4. Utilization avenues of Ash considering other chemical and physical properties:
 - a. Improvement of soil condition for agriculture purpose (Ash has soil amendment characteristics. Also Micronutrients such as Copper, Calcium, Sulphur, Zinc etc. are available).
 - b. Manufacture of Alum, Alumina, paints, ceramics, synthetic wood and other high value products like Cenosphere.
5. The characteristics of fly ash and bottom ash are different and hence the application areas for fly ash and for bottom ash are also different. Fly ash possess pozzolanic property therefore, is suitable for Manufacture of Portland Pozzolana Cement, as part replacement of cement in cement concrete and mortar, Manufacture of Ash based building products etc. Bottom ash possesses good geotechnical properties and therefore suitable for Construction of road / rail embankments, dykes, landfills, road sub-base and base course, stowing in underground mines, reclamation of de-coaled areas of open cast mines etc.
6. Storage of Ash as per Utilization Requirement

- a. In order to make ash available as per users' requirement, Fly ash and bottom ash shall be collected, transported & stored separately in lagoons. (Wherever possible)
- b. Further, storage of Bottom ash in separate lagoon is also required for using bottom ash as filter material in raising ash dyke. Provision of separate collection, transportation and storage of bottom ash & fly ash shall be explored, wherever not available.

Annexure-B

ORT Format FOR ASH Dykes

Major Safety Related Observations with Action Plan:

Sr. No.	Ash Dyke No/ Name	Date of monthly inspection done	Salient observations, specially regarding settlement, seepage, cracks, rat holes etc.	Action Taken/ Proposed Action Plan	Target Date	Responsibility	Remarks

Status of Ash Dyke with Regard to their Capacity

Sr. No.	Ash Dyke No/ Name	Last raising completed On (date)	Whether ash discharge is in progress in this dyke	Service life of ash dyke with free board Up to (date)	Planned Date of next raising	Availability of Inputs like specifications etc. from PE	Status of award of package	Completion Schedule	Responsibility

Environmental Parameters:

Sr. No.	Ash Dyke No/ Name	TSS of water effluent		Action	Whether there is any fugitive dust problem	If so, reasons thereof	Action Plan to control fugitive dust	Target Date
		going outside	going for recirculation	Plan to improve TSS				

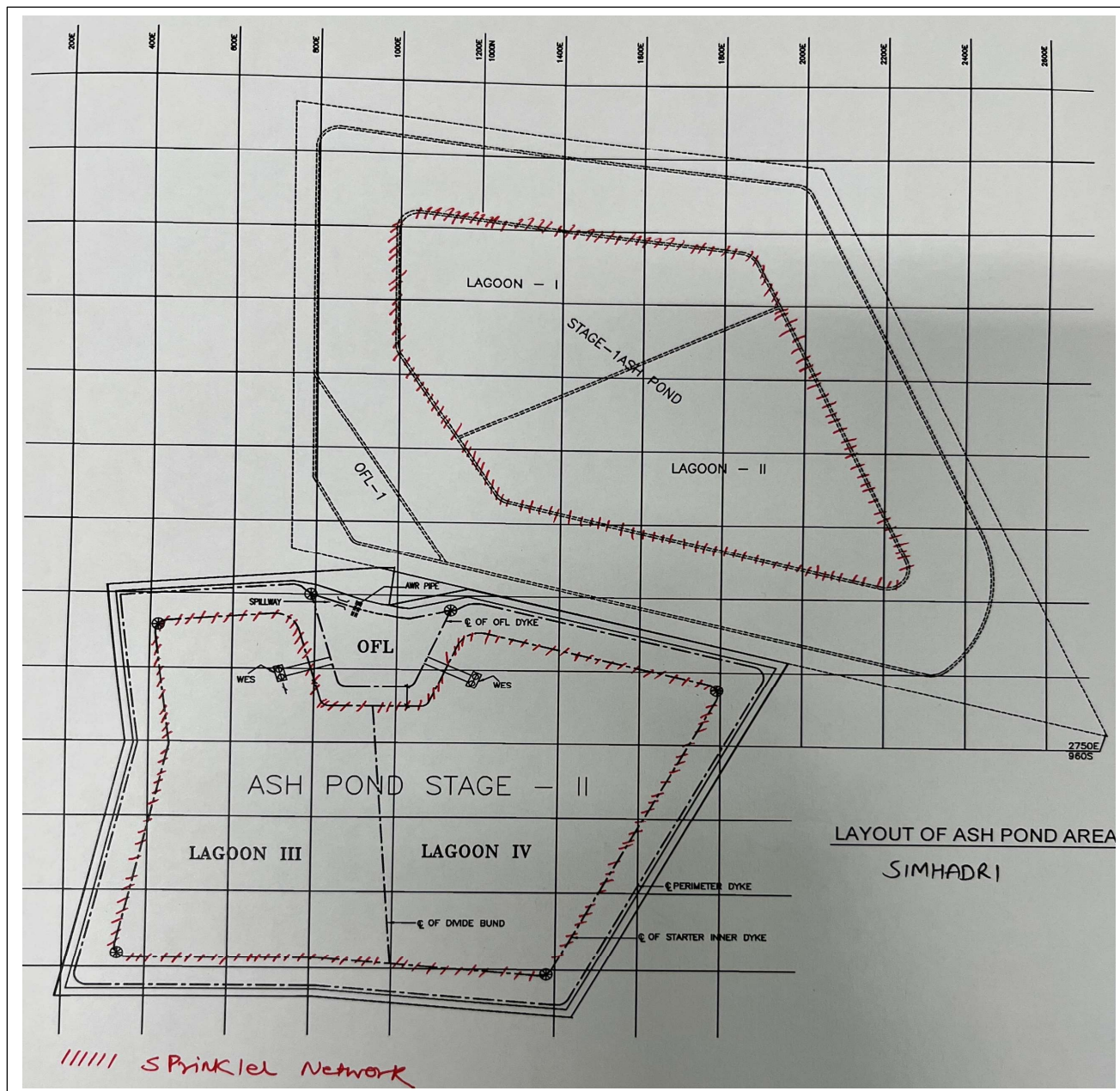
Available Free Board in the Ash Dyke (in meters):

Annexure-C

EMERGENCY PREPAREDNESS & RESPONSE PROCEDURE

SOURCE OF HAZARD	Escape of Ash Slurry into adjacent land
Area & location where it is being handled	Below the land of the ash dyke
Cause of emergency	Breach of Ash Dyke
Emergency Response	
Procedure to tackle	Inform to the Shift-in-Charge Engineer & Fire Station about the breach as soon as it is observed. - Inform higher authority about the incident. - Ask for help from Maintenance Staff & O&M – Civil dept. To stop the flow of Ash Slurry to outside the dyke. - Attend to the failed area.
Personal Safety	• First -Aid Kit should be available at, (i) Operation Control Room, (ii) CHP Control Room, (iii) First aid Centre (Plant), (iv) CISF Control Room, (v) First aid Centre (Plant) – “O”mtr. Service building. • Ambulance • Hospital: Round the clock medical aid available in township
Mitigation	Immediately stop the discharge or divert it into other ponds or lower the slurry level. If required lower the invert of water escape structure. Spilled ash around Ash Dyke shall be collected and put back into the Ash Dyke. At least 400-500 sand/ash filled bags shall be kept in readiness at site. A truck/tractor shall always be available. At any point of time, availability of local labour of 100 members should be kept in readiness in case of any emergency exists. The annual maintenance contractor/AHP contractor should also be available in case of emergency.
Immediate Measure for plugging	Plug the breach section by dumping ash bags. Once breach is plugged, the section shall be restored to original section of dyke by proper keying the new construction with existing construction.
Testing Requirement	• Round the clock inspection by ADM / Dyke group. • Mock Drill exercise (Please refer Part-I / Sl. 11)
Frequency	Yearly by Safety department in association with ADM / O&M - Civil / Ash Handling Maintenance, Operation, HR & CISF.
Record	a) Mock drill report available with Safety, Fire and respective HODs. b) Actual occurrence report available with respective HODs c) Records as per testing requirement available with respective HODs
Copy holders	DRs of Operation, Civil & AHP Maintenance, Safety & MR

Annexure-D



The sketch shows Simhadri ash dyke sprinkler network layout.

Annexure-E

Format for recording Instrument readings:

Station Name:													
Inspection Date:													
Dyke name / Lagoon name:													
Sl. No.	Start er / Raisi ng	Instrument Readings & Status											
		Piezometer							Surface Settlement Marker				
		Instru m. ID mark	C h. Mt r	T op El v. (R L)	Blank et Elevat ion (RL)		Water level obser ved (RL)	Statu s / Re mar ks	Instr um ID mark	Ch. (Mtr .)	Top Elv. (RL)		Statu s / Rem arks
					Top	Bott om					Initial	On Dat e	
1													
2													
3													

Signatures:

Annexure-F

Checklist for Monsoon Preparedness:

STATION NAME:			
Dyke Name / LAGOON Name:			Date:
Sl.	Checklist Points	Status	Remarks
1	Whether Rock-toe drains are through or any cleaning required for removal of Ash, grass, bushes, vegetation & obstructions		
2	Whether all the cross pipes are through to allow free water-flow		
3	Whether slope drains are in position with proper water-drainage path		
4	Availability of motor-able road with ramps in each level all around the dyke		
5	Whether the existing rain-cuts, if any, attended		
6	Whether Piezometer/Settlement markers are available & functioning for taking readings at close intervals during monsoon depending on requirement		
7	Whether sliding, if any, in any stretch of embankment attended		
8	Whether sand boiling areas, if any, Identified & attended with reverse filter		
9	Whether proper Turfing / grassing exists on D/s slope		
10	Whether any facility available for removal extra rainfall.		
11	Ensure regular monitoring of water level in lagoons & maintaining it to minimum possible level during monsoon		
12	Whether WES Wells and / or Spillway are functioning properly		
13	Whether WES Wells are provided with proper barricading		
14	Whether lighting mast / camera, if installed any, working properly		
Common facilities			
1	Availability of 24x7 vehicle(s) in good condition for taking rounds during monsoon.		

2	Availability of contract with scope for deployment of various machinery, e.g. Tipper, Excavator, Utility Vehicle		
3	Availability of contract with scope for deployment of adequate manpower		
4	Availability of DG set, dewatering pumps		
5	Availability of sufficient torch lights, Gum boots, Umbrellas/ raincoats		
6	Ensure availability of Aggregate, sand, earth & boulder in adequate qty. at different places, especially at probable weak zones		
7	Availability of adequate 50 kg bags filled with ash / sand / earth at different places, especially at probable weak zones		
8	Availability of empty 50 kg bags		

Annexure-G

Sample format for measures to control Fugitive emission

Station Name:												
Date:												
Sl.	Ash Dyke / Lagoon / Sub-lagoon name / Raising	Lagoon status (In Serv. / Under AU / Under construction / None)	Pond Area (LSqm)	Dust control Measures implemented in various portions with corresponding area			Remaining Exposed Ash in dry condition, if any (prone to emission)		Action Plan with target date for preceding column		Remarks, if any	
				Measures	LSqm	% of (4)	LSqm	% of (4)	Action Plan	Target		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)=(4)-(6)	(9)=(4)-(4)	(10)	(11)	(12)	
1	St# / L-1A / 1st R	In Service	6.00	Ponding	4.75	79%	0.25	4%	PO under award for 4 Water Tankers with pump & long flexible hose	07.02.22		
				Flexible sprinkler	0.50	9%	0.25	4%				Occasional controlled charging
				HDPE sheet cover	0.25	4%						
				Total	5.50	92%	0.5	8%				
2	St# / L-1B / 2 nd R	Under AU	5.00	Green curtain around AU area	3	60%	0.50	10%	PR i/p for 8 Rain Guns	07.02.22	Additionally Controlled water charging will be done in AU pits	
				Beshram plantation	1.50	30%						
				Total	4.50	90%	0.50	10%				
3	St# / L-2 / 2nd R	Under construction	4.80 (pond bed)	Beshram plantation	2.90	60%						
				Earth cover	1.90	40%						
				Total	4.80	100%	Nil	Nil				
		Embankment (u/c)	Flexible sprinkler (1 KM), Hessian Cloth with water sprinkling by tanker (20000 Sqm) & Simultaneous earth cover. Further, 0.5 KM Flexible sprinkler is being deployed by agency by 07.02.22									
Note: Stations may add other methods adopted, if any. (Like mist gun, chemical spray etc.) as applicable.												
Schematic sketch with various areas under col. (5), (6) & (8) hatched in Key plan & attached												

Annexure-H

Sample format for Ash Dyke Monthly Status Report:

- 1) Each station will send the report for each month to CC-OS-AMG by 1st working day of following month.
- 2) Each station will fill data against each point, specific to that station & delete the existing sample data while preparing the report)

Report for the Month: MM/YYYY

Report Date: DD.MM.YYYY

Station Name: TSTPS								
Sl.	Status as on Date							
A. Lagoons In service:								
Sl.	Lag Name (as applicable)	Present phase in service (Starter/Raising)	I/s wef YY	FA / BA / Mixed Ash	No. of lines in operation	Res. Cap./Life (LCM / Month)	Addition al Informati on, if any	Issues / Remark s
A1	St#I/1A	Starter	DDMMYY YY	FA (HCSD)	3	15	Cap shown is inclusive of 2* LCM AU Void as shown under SI C1	
A2	2A	4th R	DDMMYY YY	Mixed Ash	2	5		
A3								
A4								
Total Residual Capacity shown here is Excluding AU Voids under SI. C (Unless Noted Otherwise):						20 LMT	20 LMT includes 2 *LMT AU Void in Lag-1A but is excluding 3 LMT AU void in Lag-5A	
B. Lagoons under construction (Envisaged in FR):								
Sl.	Lag Name	Existing Raising	U/c wef	Present construc tion / raising / buttressi ng	Physical Progres s in %	Target (Charging Readiness Date / Envisaged Capacity LCM)	Addition al Informati on, if any	Issues / Remark s

B1	3A	3rd	DDMMYY YY	4th Raising / Buttressing PO No. ...	80%	DDMMYYY Y / 6 LCM	Ash work done 100 %, E/cover done 70%	Garlandi ng i/p
B2	4A	4th	DDMMYY YY	Raising / Buttressin g PO No. ...	15%	Progressiv ely from DDMMYY YY / 4 LCM	Ash work done 25 %, E/cover done 5%	
B3								
B4								
Station Name: TSTPS								
Sl.	Status as on Date							
C. Lagoons under AU:								
Sl.	Lag Name	AU i/p from Raising	Under A U wef	Res voi d (LCM)	AU plan till	Charging plan wef	Addition al Informati on, if any	Issues / Remark s
C1	1A	Starter	DDMMYY YY	2*		AU pit is already i n service		Lag-1A is both under AU & in service
C2	5A	5th R	DDMMYY YY	3				
C3								
D. Capacity enhancement (Beyond FR) status: Please provide action taken / PR/PO nos. with present status as applicable								
Sl.	Lag Name	Propo sed Raisin g	Engg.	Investment approval	Raising PR / PO No. / Status	Targe t (Awar d)	Target (Charging Readines s) / Envisage d Capacity LCM	Issues / Remark s
D1	1A	1st R	Drg. received	NA	PR No. ...	DDMMYYY Y	DDMMYYY Y / xxx LCM	NIT bein g floated
D2	2A	Buttressin g	Consultan cy awarded (PO No.	Initiation target: DDMMYYY Y	Initiation Target DDMMY YYY	DDMMYYY Y	DDMMYYY Y / xxx LCM	Prel. Report from IIT by DDMMY

			...) Target: DDMMYY YY					YYY
E. Stability Study Due as per OGN for Lagoon (names / No./Raising) with Status: PR/PO No....								
Lag Name	Phase at present (Construction Completed) (Starter/Raising)	Stability Study Status						
		Last Done (DDMMYYYY)	Next Due Plan	Status	Remarks			
F. Annual Statutory Certification: FY:20YY-YY: Action planned / taken along with present status								
G. Pending Action Points of Last Audit – Status with Action Plan for compliance:								
Pending Action Points of Last Audit Present Status & Revised target								
Sl.	Action Point	Target as per Report	Present Status	Revised Target	Issues/ Remarks, if any			
H. OFL Dredging, Seepage water pump house, etc								
I. Mock Drill status:								
J. Any other issue / exceptions / distress / development / information, if any								

Annexure-I

Guidelines for recovering land from existing Ash Ponds of NTPC Stations



NTPC Limited

1.0 Objective and Scope

As per existing MoEF notification (3rd November 2009), land area is allowed @ 0.25 Acre per MW of plant capacity for emergency ash pond for ash storage. As per MoEF, the target fly ash utilization is 50%, 70%, 90% & 100% in 1st year, 2nd year, 3rd year & 4th year respectively from the date of commissioning of power plant. The balance unutilized ash accumulated during first four years (the difference between generation and utilization target) shall be utilized progressively over next five years in addition to 100% utilization of current generation of fly ash.

Based on present rate of ash utilization of about more than 100% in some NTPC Stations and exhausting of the above period allowed by MOEF for emergency disposal of unutilized ash in ashpond, a guideline is prepared with the objective to recover land of Ash pond from the Stations in a sustainable manner. The proposed guideline is limited to safely evacuation of the deposited ash from inside the storage lagoons and dismantling the associated structures such as dykes/WES wells/ Spillways etc. (excluding Pump houses); however the utilization/ disposal of the evacuated materials shall be strategically decided by Site in eco-friendly manner.

2.0 Ash Pond areas

The ash generated from the power plants is disposed-off in the Ash ponds. Unlike water reservoir, the dyke embankments for ash pond are generally not constructed up to ultimate height in one go and initially constructed up to a limited height with provision of subsequent raising as per requirement. Ash pond is divided into lagoons and provided with garlanding arrangements for changeover of the ash slurry feed points for even filling of the pond and for effective settlement of the ash particles. A photograph of actual ash storage lagoon of NTPC with ash slurry discharge line is presented in Fig.1.



Fig.1. Photograph of actual ash pond of NTPC with ash discharge line

Ash pond is designed as multi-lagoon systems with minimum two storage lagoons and one over- flow lagoon (OFL). Having two or more storage lagoon facilitates sequential raising of lagoons by putting one lagoon for ash filling while the other lagoon is used for raising its dyke. OFL helps in controlling the effluent quality of the supernatant, which is recycled back to plant for making ash slurry. The typical layout and cross-section of storage lagoon and OFL is shown in the following Fig.2 and Fig.3 respectively.

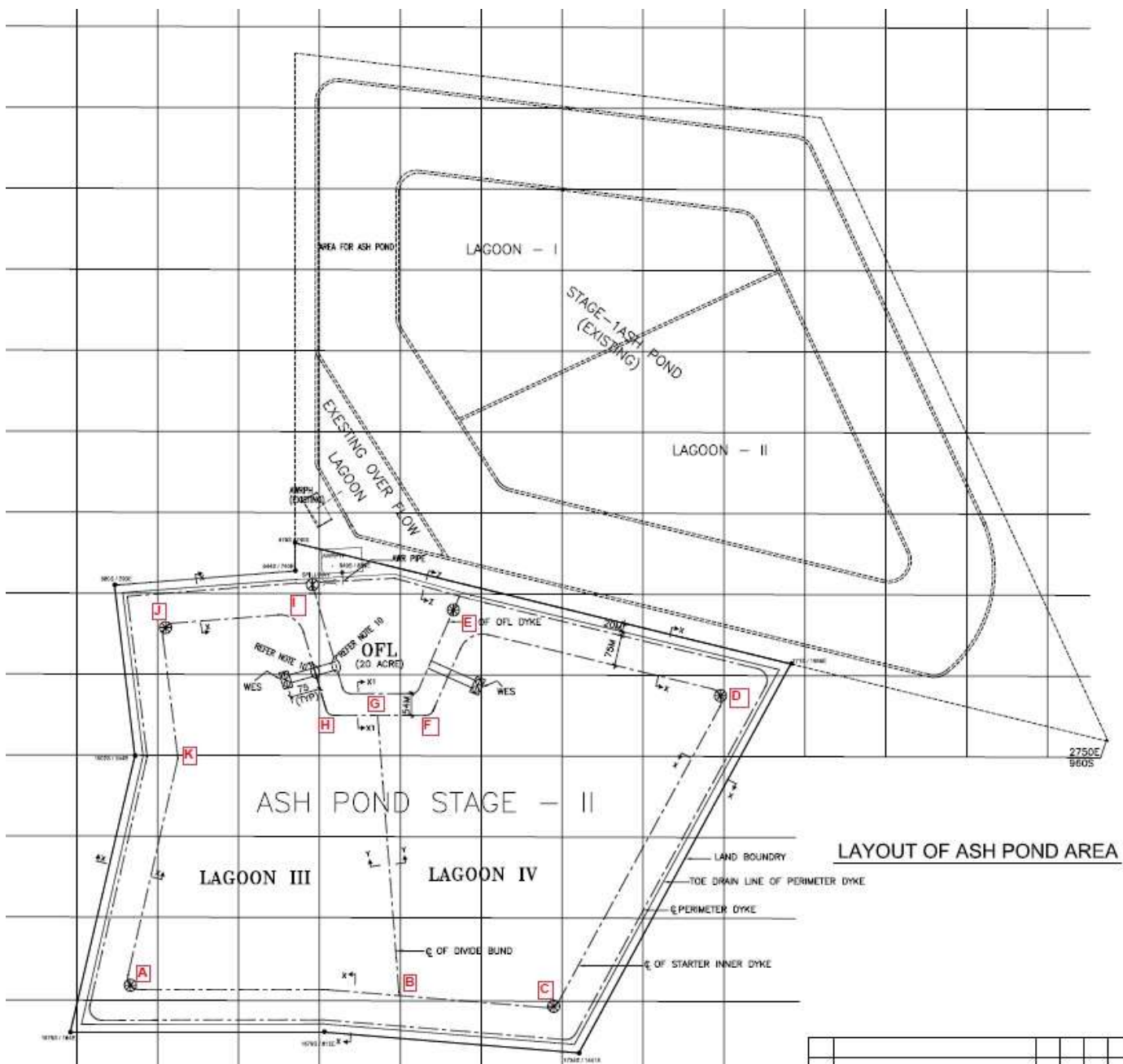


Fig.2: Typical layout of ash pond

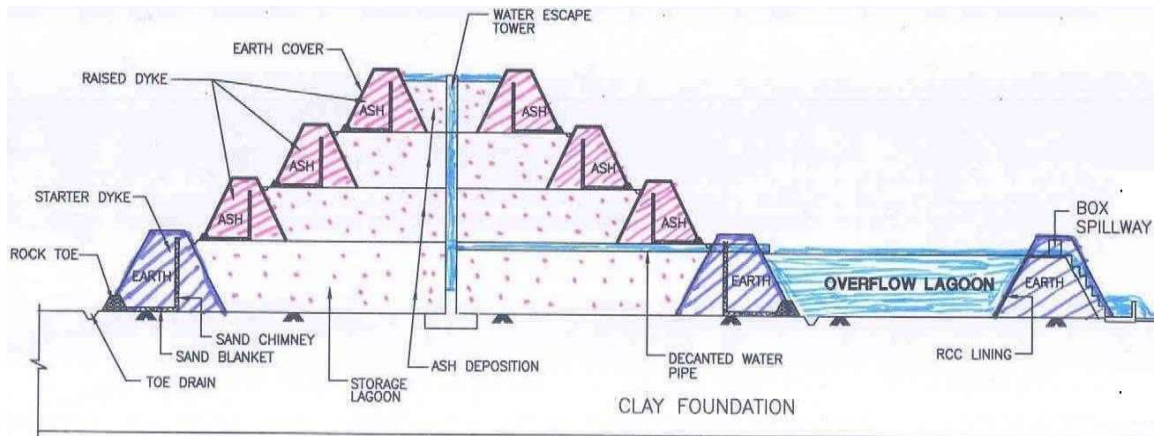


Fig. 3. Typical cross-section of storage lagoon and OFL

3.0 Reclaiming Land

3.1 Reclaiming Land from abandoned Ash Pond

3.1.1 Evacuation of Ash from inside Ash Pond

The ash dykes are stable and safe with design factor of safety as per Indian standard codes. As the ash dykes are constructed up to ultimate heights in different stages; the evacuation of ash from storage lagoons shall be done in phases/ stages. Parallely, dismantling of associated structures shall be carried out in stages with proper care. A typical section of storage lagoon with ultimate height (considering maximum four raisings) is presented following in Figure.4.

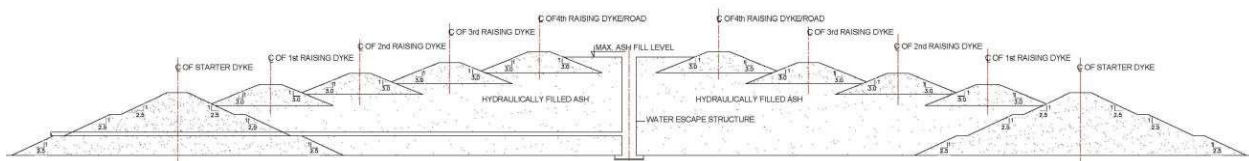


Fig.4: Typ. section of storage lagoon with ultimate height (with max. four raisings)

3.1.2 From Raising dykes

The raising dykes are constructed using ash and number of raisings are depending on maximum height of Starter dyke, capacity requirement and foundation conditions. As the raising was done in stages; the evacuation of ash

from each raising shall also be carried out in stages. Before taking up evacuation from the related areas, all the plants/ trees/ bushes etc shall be safely removed.

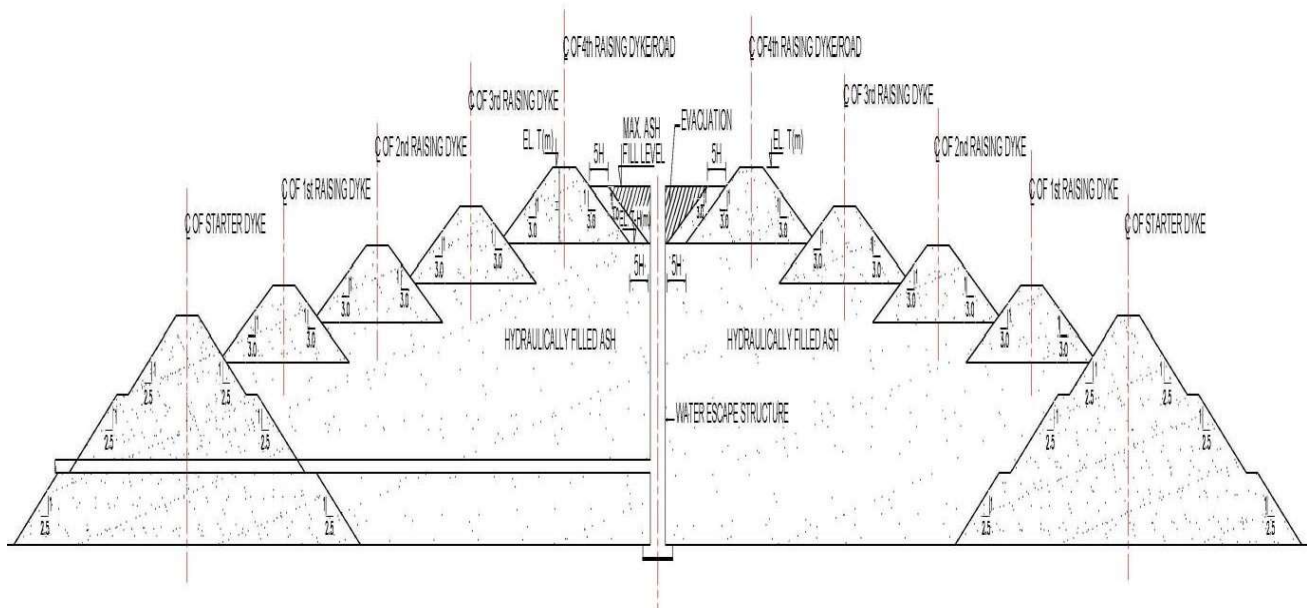
Further, during ash evacuation, all the necessary precautions shall be taken by Site against water and environmental pollution, and where required undertake the additional works to achieve this objective.

The Water escape wells shall remain functional and the closed flexible opening initially provided may be progressively opened to allow the escape of surface run-off through WES wells for recirculation. WES wells shall be dismantled in subsequent stages after evacuation of dyke raisings of each stage. The evacuation of the same shall be carried out in two phases as given below to ensure safety.

Phase-1: Initially, ash evacuation should start from safe distance (five times the height of existing top most dyke) away from heel of the dyke to ensure the dyke stability and safety. The depth of excavation shall be limited to ash bed level of the existing top most raising dyke. In any case, the slope of excavated surface shall not be steeper than 1(V):3.0(H) and no berms are required for the excavated height of 6.0m.

Phase-2: The raised dyke and balance part of deposited ash kept in phase-I (for maintaining the safe distance of excavation) may be evacuated and subsequently the WES Wells may be dismantled with proper surface drainage. The dismantled height of WES wells shall be limited to 300mm above the excavated ash level to avoid any ash movement in WES wells.

Wherever there are Spillways, they shall be separately dismantled and disposed off in eco- friendly manner before evacuation of dyke portion. The functionality of rock-toe shall be ensured till the evacuation stage of related dyke. The sequence of ash evacuation and dismantling of associated structures (dyke embankment & WES wells) shall be in accordance with the phases/ stages as mentioned in the



following figures.

Fig.5 (Phase-1): Evacuation of ash from top most raising

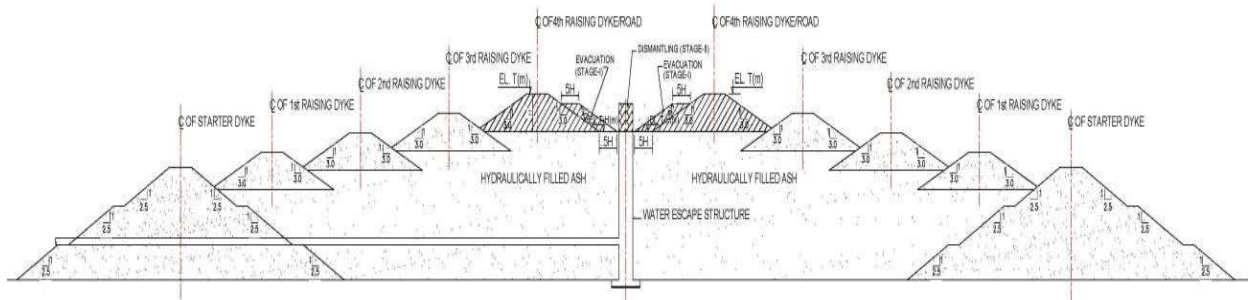


Fig.6(Phase-2): Evacuation of topmost raising dyke with adjoining area (Stage-I) and Dismantling of WES well (Stage-II)

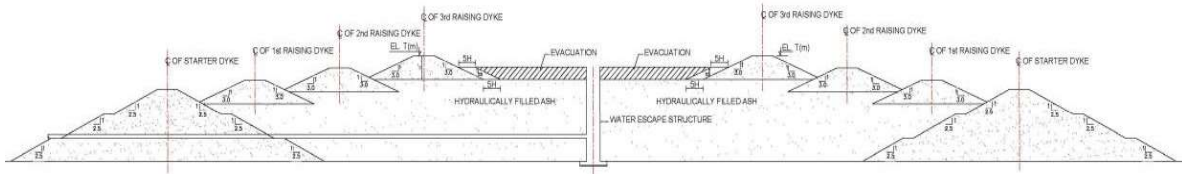


Fig.7(Phase-1): Evacuation of ash from 3rd raising

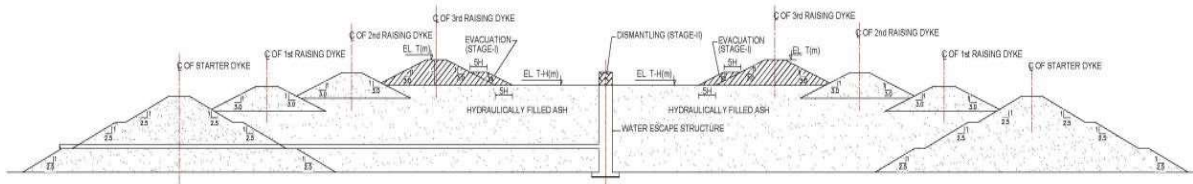


Fig.8 (Phase-2): Evacuation of 3rd raising dyke with adjoining area (Stage-I) and Dismantling of WES well (Stage-II)

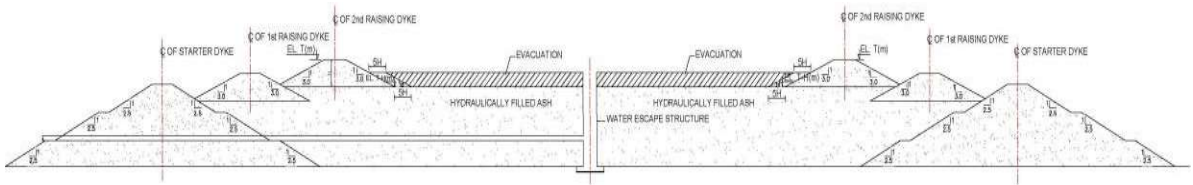


Fig.9 (Phase-1): Evacuation of ash from 2nd raising

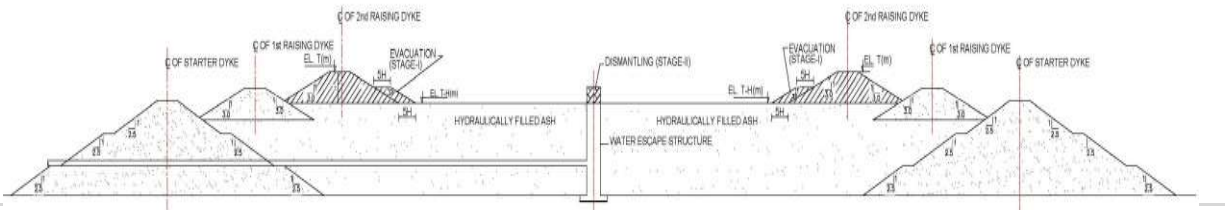


Fig.10(Phase-2): Evacuation of 2nd raising dyke with adjoining area (Stage-I) and Dismantling of WES well (Stage-II)

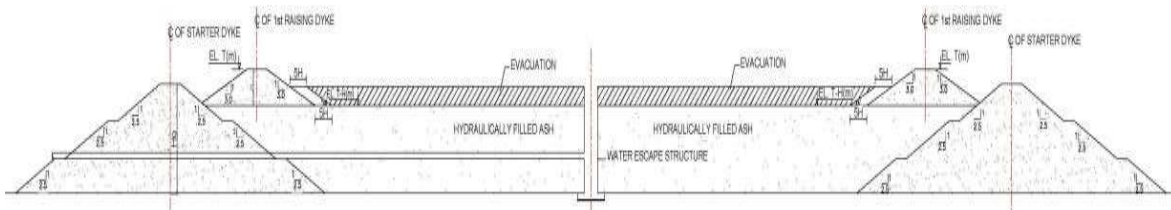


Fig.11(Phase-1): Evacuation of ash from 1st raising

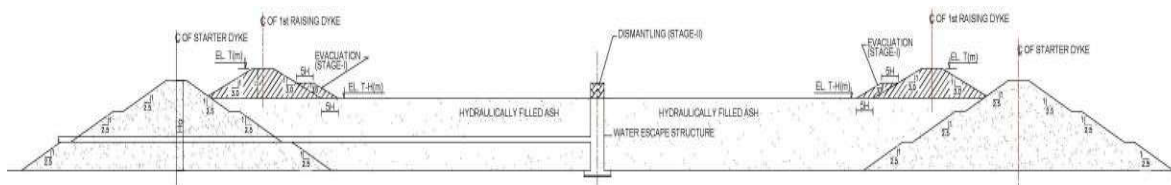


Fig.12(Phase-2): Evacuation of 1st raising dyke with adjoining area (Stage-I) and Dismantling of WES well (Stage-II)

3.1.3 From Starter dyke

The Starter dykes are constructed using earth and there are hume pipes which are embedded in Starter dyke embankment. Before taking up evacuation from the related areas, all the plants/ trees/ bushes etc. shall be removed. The maximum height of Starter dyke is varying depending on capacity requirement and foundation conditions. Accordingly, the evacuation of the same shall be carried out in three phases as mentioned below to ensure safety.

Phase-1: Ash evacuation should start from safe distance (five times the height of existing starter dyke) away from heel of the dyke to ensure the dyke stability and safety. The depth of excavation shall be up to natural bed level excluding the approach road of WES wells (up to top of WES pipes). In any case, the slope of excavated surface shall not be steeper than 1(V):2.5(H) and 3.0m wide berms are required to be provided at every 6.0m from top of starter dyke. Typical arrangement for the same is shown in following Fig.13.

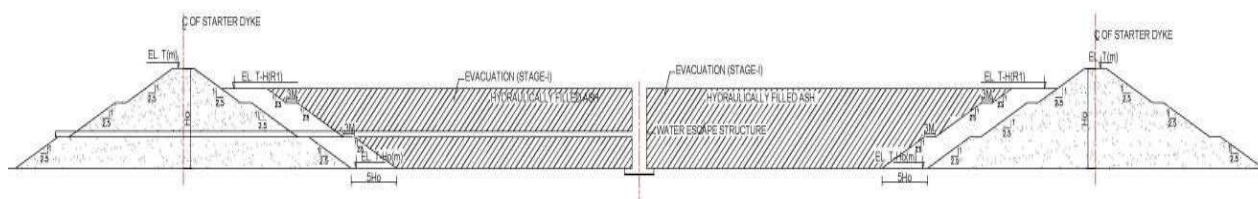


Fig.13(Phase-1): Evacuation of ash from Starter dyke

Phase-2: The Starter dyke embankment and balance part of deposited ash kept in above phase- I (for maintaining the safe distance of excavation) may be evacuated and subsequently the WES Wells may be dismantled up to encased level of Hume pipes. The dismantled height of WES wells shall be limited to 300mm above the excavated ash level to avoid any ash movement in WES wells. Typical arrangement for the same is shown in following Fig.14.

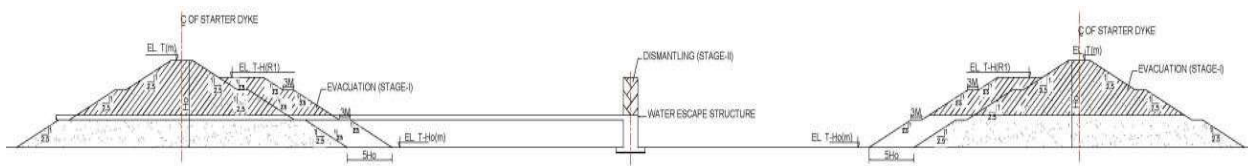


Fig.14(Phase-2): Evacuation of adjoining area of Starter dyke(Stage-I)

and Dismantling of WES well up to top level of hume pipe

(Stage-II)

Phase-3: The balance part of Starter dyke embankment, adjoining deposited ash may be evacuated up to natural bed level and WES Wells along with its approach road may be dismantled with suitable arrangement of surface drainage to avoid water logging in the reclaimed areas. Typical arrangement for the same is shown in following Fig.15.

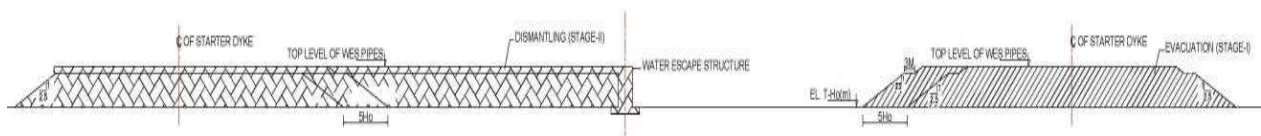


Fig.15(Phase-3): Evacuation of Starter dyke (Stage-I) and Dismantling of

Balance part of WES well (Stage-II)

3.2 Reclaiming Land from operational Ash Pond

3.2.1 Operational Ash Pond

Ash disposal areas are designed as multi-lagoon systems with minimum two

storage lagoons and one over-flow lagoon to meet functional and operational requirements. Further, ash slurry pipeline with arrangements for changeover of the ash slurry feed points are done by Site based on operational requirements and prevailing site conditions. To reclaim the land from operational ash pond, detailed planning needs to be done by Site to ensure sustaining the power generation with enhanced ash utilization.

As a 1st step for reclaiming land, the efforts may be made by Stations to go for enhanced ash utilisation in alternate to the construction of new dyke/ raising the existing dykes/ buttressing etc. Further, one of the operational lagoon may be abandoned for ash slurry discharge and decanted water in the storage lagoon may be allowed to escape through WES wells. Further, the abandoned lagoon may be planned for taking up the ash evacuation with proper surface drainage arrangements. The sequence of evacuation and dismantling of the associated structures shall be in accordance of the phases/ stages as mentioned in above paras 3.6.1.3 & 3.6.1.4 for planning of recovering land from one of the abandoned lagoon. Subsequently, after recovering the total land from this lagoon, the other operative lagoon may be planned for abandoning from ash slurry discharge and the above steps of evacuation and dismantling of associated structures may be followed.

Further, during ash evacuation, all the necessary precautions shall be taken by Site against water and environmental pollution, and where required undertake the additional works to achieve this objective. The rock-toe shall remain functional till the evacuation stage of related dyke. The Water escape wells shall remain functional and the closed flexible opening initially provided may be progressively opened to allow the escape of surface run-off through WES wells for recirculation.

3.2.2 Reclaiming Land from Overflow Lagoon

The Overflow lagoon (OFL) shall be maintained till the evacuation of main Starter dykes of storage lagoon is safely completed.

Further, most of the water of the OFL shall be initially recirculated back to plant using existing AWRS system using evacuation pipe embedded in OFL dyke. Efforts may also be made to recirculate remaining water below the invert level of evacuation pipe embedded in dyke body with suitable arrangements. If required additional dewatering pumps may also be installed for the same.

Subsequent to the complete evacuation of OFL water, the dyke body shall be evacuated in eco- friendly manner. All the plants/ trees/ bushes etc shall be removed from the working area of ashpond before proceeding for ash evacuation from storage lagoons.

After reclaiming the land from ash pond area with safe evacuation of deposited ash, dyke embankments etc., suitable arrangement of surface drainage to be provided as per actual site condition to avoid water logging in the reclaimed areas.

3.3 Vehicle movement over Dykes

For transportation of evacuated ash/ dismantled materials etc., the inspection road constructed at ground level (away from toe drain) shall be used for regular plying of heavily loaded vehicles and movement of heavily loaded vehicles over ash dyke embankment should be avoided; however exit/ entry of the vehicles into or from ash pond may done by Site by providing additional number of suitable approaches/ ramps on either side as per requirement and site condition.

3.4 Precautions during evacuation of ash from ash pond area

During ash evacuation, all the necessary precautions shall be taken up by Site against ground water contamination and environmental pollution, and wherever required, undertake the additional works to achieve this objective. The Water escape wells shall remain functional and the closed flexible opening initially provided may be progressively opened to allow the escape of surface run-off through WES wells for recirculation back to plant. Wherever there are Spillways, they shall be separately dismantled and disposed off in eco-friendly manner

before evacuation of dyke portion. The sequence of ash evacuation and dismantling of associated structures (dyke embankment & WES wells) shall be in accordance with the phases/ stages as mentioned in the above respective figures. A Typical sketch for safe excavation of ash from inside ash pond is also presented below

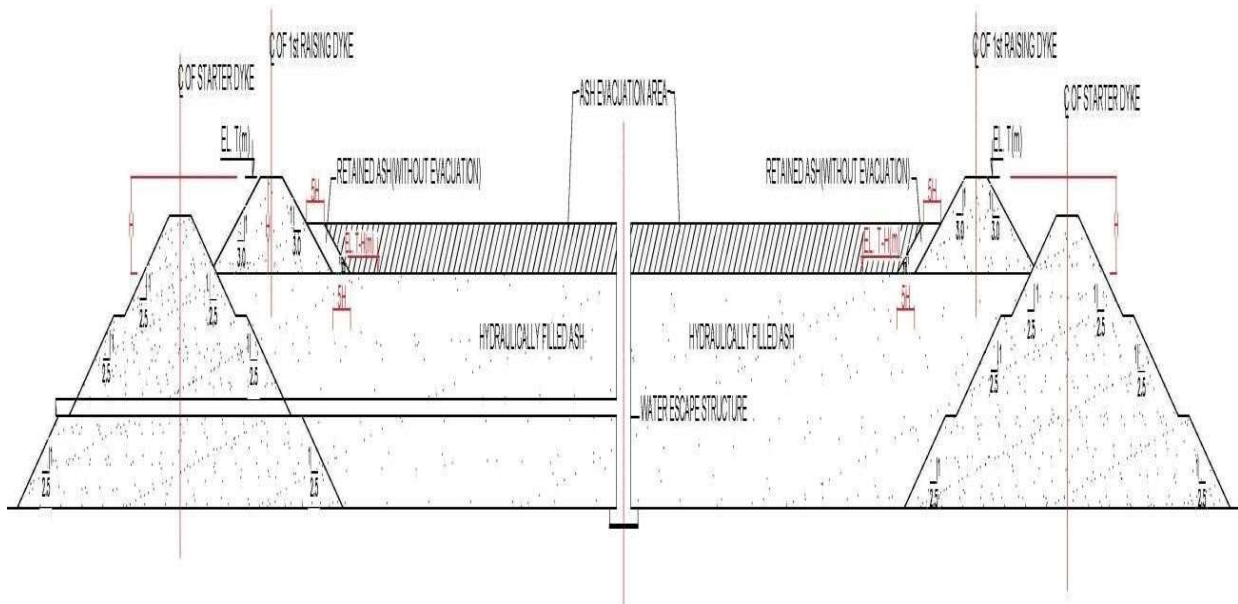
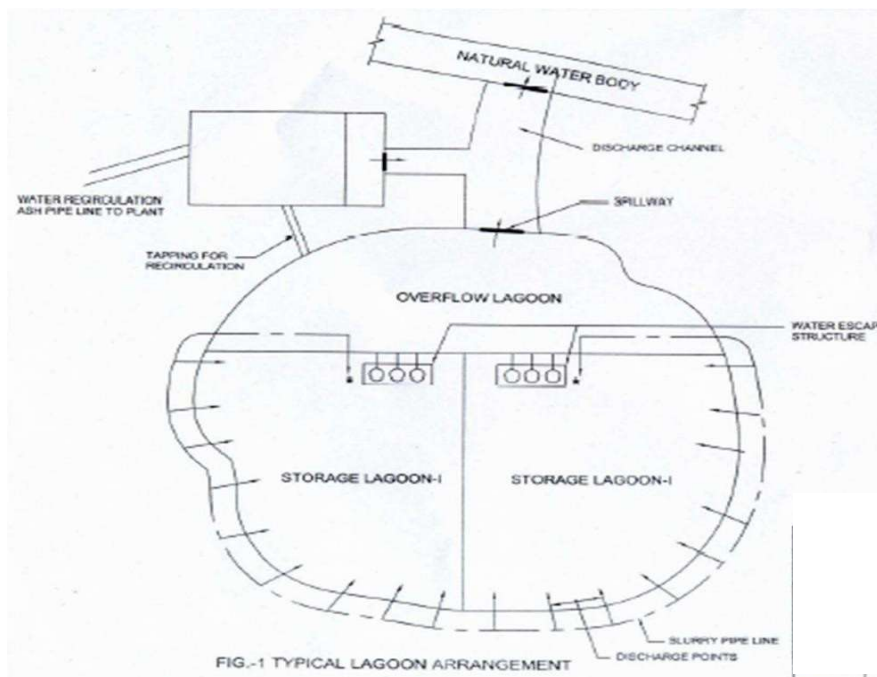


Fig: Typical arrangement for safe excavation of ash from inside ash pond

Annexure-J : Schematic sketches showing typical lagoon arrangement & WES

Layout of Ash Dyke, without HCSD



Ash Filling shall be carried out near the water escape in the initial periods

Annexure-K: IOM dated 16th Aug 2023, issued by ED (OS) to ensure healthiness of WES



CORPORATE OPERATION SERVICES

INTER OFFICE MEMORANDUM

From: ED(OS)	To: All HOPs, Coal Stations
Copy (For information)	1. All REDs 2. STA to Dir (O)
Date	16 August 2023
Ref No.	COS/IOM/ED/22/064
Subject: Actions to be taken for Ash Dyke WES – RCC Decanting Wells & Hume Pipes	

FO & PL is being witnessed in one of our stations due to recent development of issues related to WES(Water Escape Structure) wells & RCC decanting pipes in Ash Dyke.

Accordingly, all stations operating with ash dyke facility are requested to take following actions:

1. After exhaustion of any particular stage (raising) of any lagoon and taking up construction of vertical extension of Wells for subsequent raising, following points are to be adhered:
 - a) Inspection of well slits to be done to ensure proper closing (Drone inspection may be considered).
 - b) Outflow from RCC decanting pipes shall be checked to get an idea about probability of any choking or flow restriction due to obstruction / ash deposition inside well / decant pipe. In case of any deviation, the same shall be referred to CC-Engg & CC-OS. Robotic inspection may be considered if the situation warrants.
2. Construction of Wells shall be carried out strictly as per latest RFC drawings issued by CC-Engg, including size & quality of precast slabs.
3. After completion of vertical extension of WES, closure of the flexible opening in previous raising with in-situ RCC shall be ensured.
4. Mesh of 25mm dia. Rebars on top of well in previous raising shall be removed after a new mesh is provided on top of extended well.

The above shall be implemented with immediate effect.



ARINDAM SINHA



से विद्युत विकास में समर्पित

**NTPC SIMHADRI, GUIDELINES FOR ASH DYKE
MANAGEMENT LMI/OGN/OPS/SYST/022
REV. 5, March 2025**

**सिम्हाद्री
SIMHADRI**

Annexure-L: IOM dated 9th Dec 2024, issued by ED (OS) to ensure safety of men & equipment involved in heavy vehicle movement at Ash Dyke Area



CORPORATE OPERATION SERVICES

INTER OFFICE MEMORANDUM

From: ED (OS)	To: All GM (O&M)s, Coal Stations
Copy (For information)	1. All REDs 2. STA to Dir (O)
Date	09 December 2024
Ref No.	COS/ED/AMG/2024/53

SUBJECT: Actions to be taken for safety of men & equipment involved in heavy vehicle movement at Ash Dyke Area

A fatal incident has recently taken place in one of our stations during pond ash utilization from Ash Dyke area, when one loaded vehicle overturned while coming down from 2nd raising top through a ramp to the lower level. Substantial movement of heavy vehicular traffic at the ash dyke area is taking place at our stations to meet their respective ash utilization targets, that highly warrants the requirement of having proper & safe ramps at ash dyke to ensure safety of men & equipment engaged and smooth traffic movement.

Accordingly, all stations operating with ash dyke facility are requested to take following actions:

1. Heavy vehicular movement at ash dyke shall be preplanned with route(s) predefined and approved by P&S. Movement through any other route shall be strictly prohibited.
2. Exit/ entry of the heavy vehicles into or from ash pond shall be done by Site by constructing ramps, with Bituminous concrete topping, on upstream & downstream sides of the embankment as per requirement and site condition. For details of ramp, the indicative drawing in enclosed Annexure -1 may be referred to.
3. The ramp shall be widened to 10 meters at the junctions where it meets the embankment top for smooth turning of the trucks.
4. Provision of guard walls shall also be made along the outer edge on either side of the ramps with fluorescent strips in alternate white & black color.
5. Caution signboards like 'Speed limit 20 Kmph' as per NTPC policy, 'No overtaking' etc. shall be displayed all along the route of heavy vehicular movement.
6. Deployment of traffic controllers may be done by the ash lifting agencies based on site specific requirements.
7. Breath Analyzers shall be purchased & used to prevent entry of drunk truck drivers engaged in ash dyke area. Records shall be maintained by Site AUD.
8. Flood light masts and CC Cameras shall be installed at various locations as required for area illumination and monitoring the dyke area as per OGN, ref. no. OGN-OPS-SYST-022 Rev. No. 3, for Guidelines for Ash Dyke Management in NTPC Stations.

The above shall be implemented with immediate effect. ATR with target dates shall be provided to AGM, COS-AMG by 16th December 2024.

Encl: Ann-1

A.K. Manohar
(A K MANOHAR) 09/12/2024
Page 1 | 1

Annexure-L

Attachment to IOM dated 9th Dec 2024, issued by ED (OS) to ensure safety of men & equipment involved in heavy vehicle movement at Ash Dyke Area

